Renal Physiology: Regulation of Sodium and Water Excretion

(How the kidney defends blood volume and osmolality)

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College of Osteopathic Medicine

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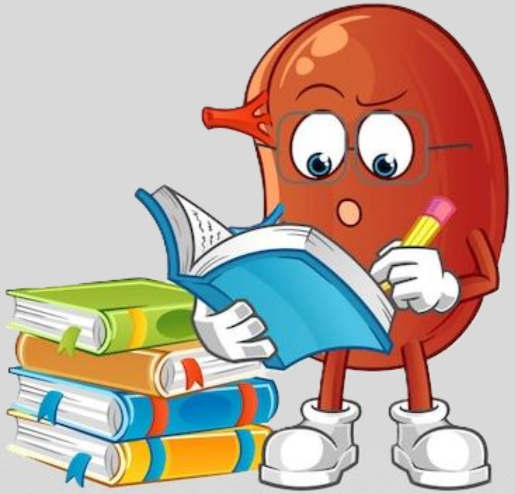

Learning Objectives

1. **Differentiate** how the kidney independently regulates extracellular fluid **volume** (via sodium excretion) and plasma **osmolality/serum [Na^+]** (via water excretion), and **explain** why "sodium sets volume, water sets concentration" is the organizing principle of renal fluid balance.

2. **Trace** the renin-angiotensin-aldosterone system as a turn-based, self-limiting cascade from the three renin triggers through angiotensin II and aldosterone to their effects on vascular tone and sodium reabsorption

3. **Analyze** how osmoreceptors and baroreceptors jointly govern ADH secretion and **justify** why a dangerous fall in effective circulating volume overrides osmoregulation.

4. **Distinguish the three systems of blood-pressure control by their time course and mechanism** and **explain** why each covers the timescale the previous one cannot.

5. **Integrate** renal and cardiovascular mechanisms to **diagnose** why the kidney's normal volume-defense response becomes maladaptive in congestive heart failure.

Resources Used

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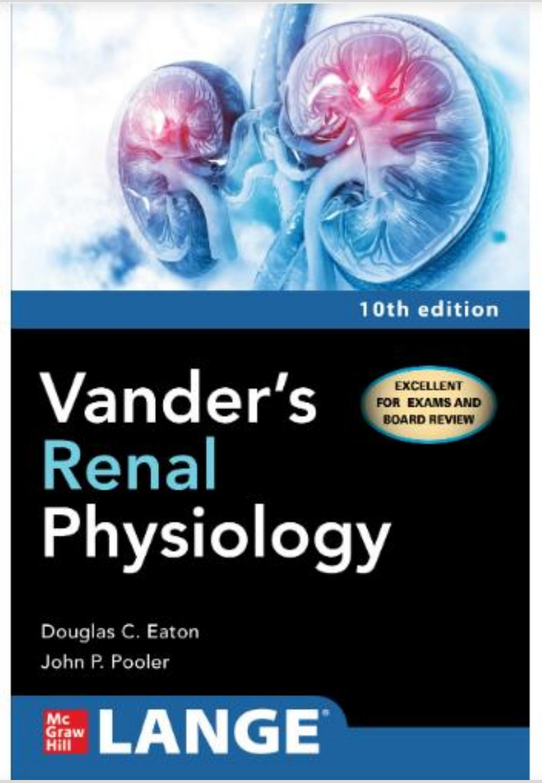

Chapters 6 and 7. Available free online through NYIT Medical Library. From Medical Library Website, go to Find and Request → Search Books and Media → Basic Search "Vander's Renal Physiology"
**Textbook provides practice questions!!!

UWorld
YOUR WORLD OF LEARNING

(Correlating to Boards relevancy)

Lecture Format

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Part 1: Where sodium and water move & why it matters

Part 2: How the body regulates those movements

Part 3: Why dysregulation causes disease

Opening Question:

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A 75-year-old male with PMHx of congestive heart failure presents with complaints of dyspnea. Physical exam reveals 3+ bilateral lower extremity pitting edema (excess fluid in legs) and crackles at the bilateral lung base due to pulmonary edema (fluid buildup in lungs).

The physician mentions that his body is clearly overloaded with fluid, and CMP displays low blood sodium (128 mEq/L) (Normal = 135-145 mEq/L). The physician mentions his kidneys are avidly retaining even more salt and water, making the swelling worse.

1) If this patient is fluid overloaded, why are his kidneys acting like he's dehydrated (retaining more salt and water)?
2) If he's retaining salt and water, why is his sodium low?
3) Why do we treat him with drugs that act on the kidney for a heart problem?

Body Fluid Compartments (Review)

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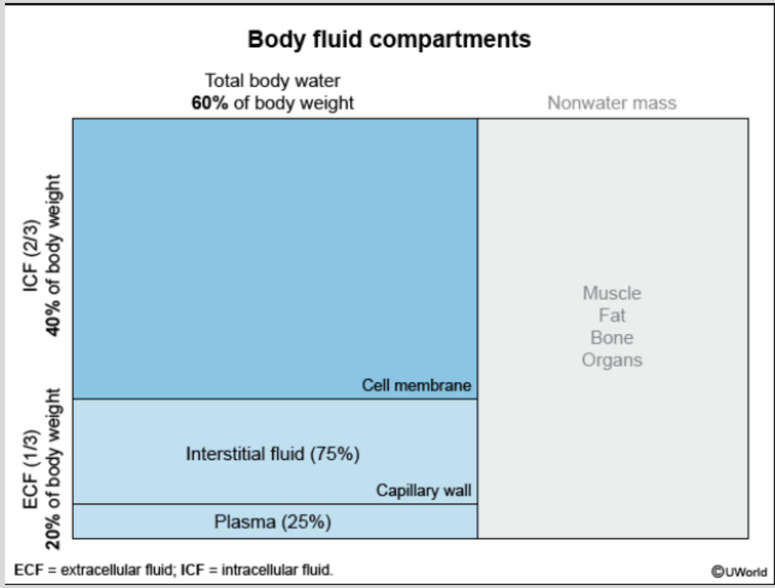
- 60% of body weight is water
  - 2/3 of that 60% is **intracellular fluid (ICF)** = the fluid in cells
  - 1/3 of that is **extracellular fluid (ECF)** → further broken down into **interstitial fluid** (3/4 of ECF) + **blood plasma** (1/4 of ECF) (fraction of % is transcellular = CSF, synovial fluid)
    - Interstitial fluid surrounds cells, providing a medium for nutrient /waste exchange
    - **Blood Plasma = the fluid compartment of blood** and 55% of blood volume (remaining 45% of blood volume = erythrocytes)
    - Capillary walls separate plasma and interstitial fluid, and fluid exchange between this barrier are governed by Starling forces
  - **ECF and ICF are in osmotic equilibrium**; **the shift of water between/within these compartments depends on the gain and loss of <u>sodium</u>** **(and other solutes)**
    - Because of cell Na-K-ATPases, intracellular sodium concentrations are fairly stable; therefore, additions/losses of sodium from the body are **mostly to/from the ECF**
    - **Sodium = major ECF osmole**
    - ***<u>Where sodium/the osmoles move, water follows</u>***



Osmolarity/Osmolality = total solute concentration

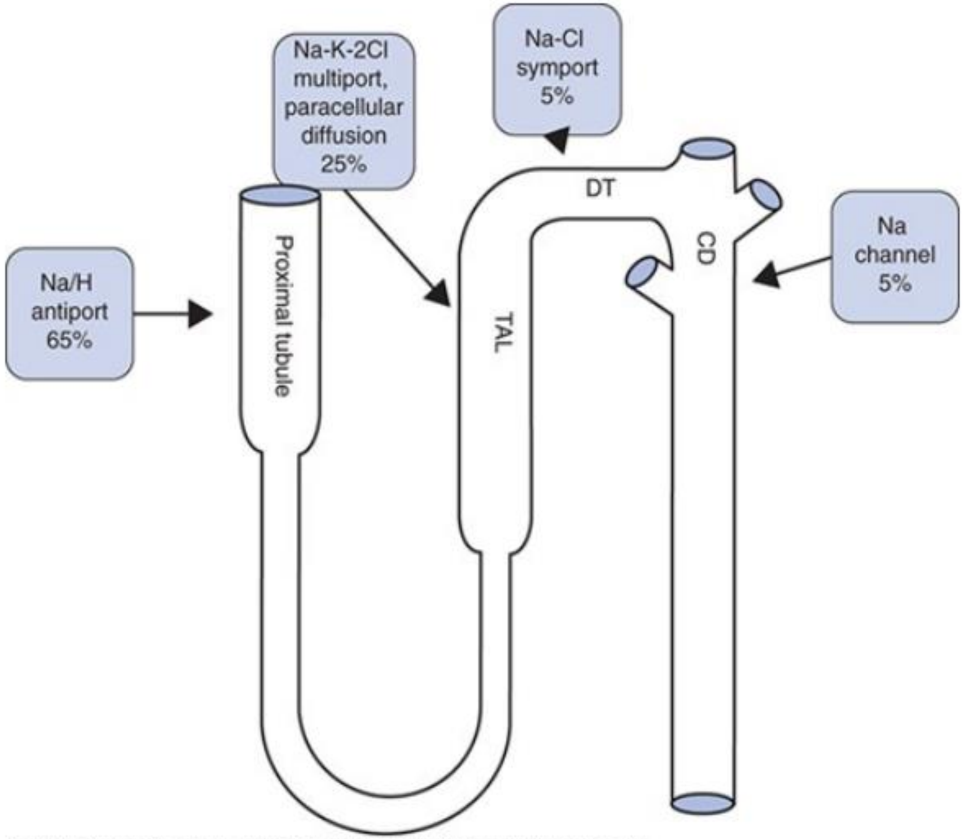
Tonicity = effect of solutes on water movement across membranes

Reabsorption of Sodium Along Nephron

1) Sodium starts in tubular lumen (if it stays here, it will be excreted)
2) Sodium crosses tubular wall **ACTIVELY**, via **apical** transporters (*transporter depends on segment*)
   - **Proximal tubule**: NHE3 (Na/H antiport), SGLT (Sodium glucose symporter), Na-aa (BULK ISOOSMOTIC TRANSPORT)
   - **Descending limb**: NO REABSORPTION OF SODIUM
   - **Thick Ascending Limb**: **NKCC2 (Na-K-2Cl Multiporter)**
     - **diluting segment – salt leaves, water stays**
     - Loop diuretics target
   - **Distal Convoluted Tubule**: **NCC (Na-Cl Symporter)**
     - Thiazide diuretics target
   - **Collecting Duct (Principal Cell)**: **ENaC - fine-tuned by ALDOSTERONE**— *key hormone of sodium/volume balance*

3) Sodium exits cell into interstitium via **basolateral Na-K-ATPases (active transport)**

4) Sodium enters nearby capillaries (peritubular in cortex, vasa recta in medulla) → reclaimed into bloodstream



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

Tubular lumen
↓ (apical membrane)
Tubular epithelial cell
↓ (basolateral Na-K-ATPase)
Renal interstitium
↓
Peritubular capillary / vasa recta
↓
Bloodstream

Reabsorption of Water Along Nephron

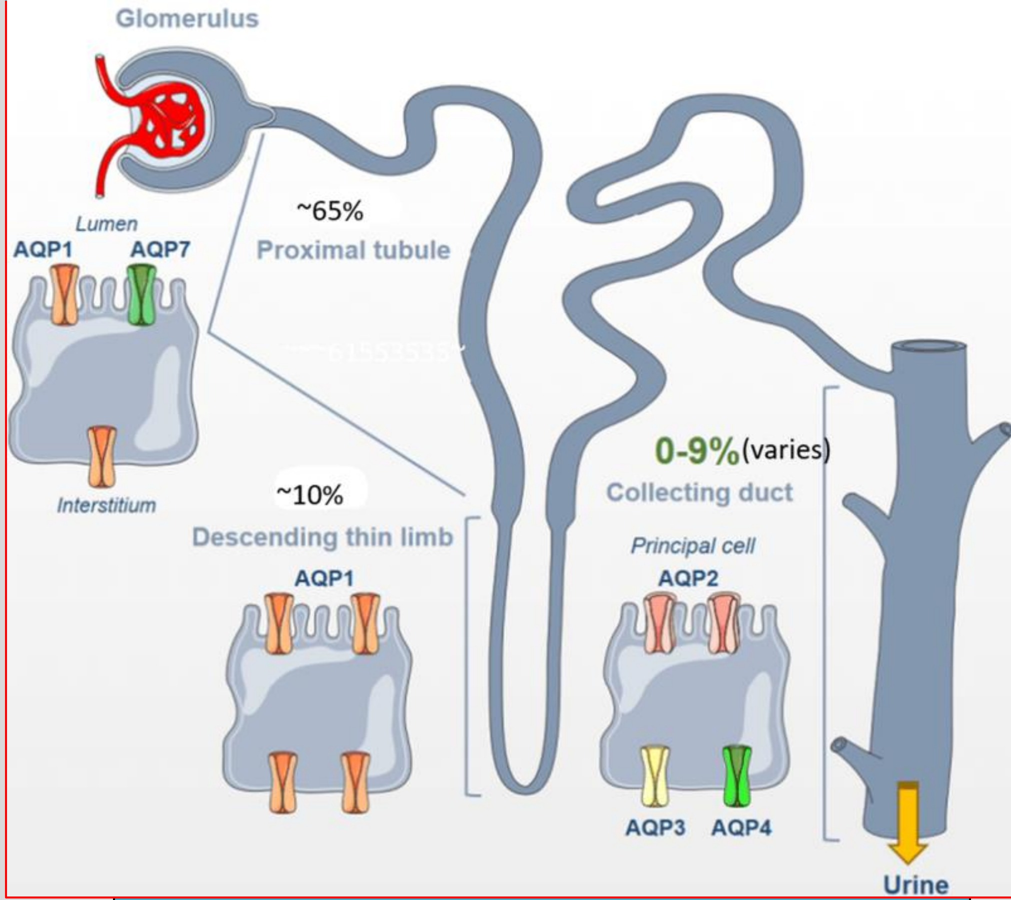
1) Water starts in the tubular lumen (if it stays here, it will be excreted)

2) Water crosses the tubule wall PASSIVELY, by osmosis — only where the wall is water-permeable (*permeability depends on segment*)

- **Proximal tubule:** freely permeable (**AQP1**) — ~65%, reabsorbed **isosmotically** (follows Na^+/solute)
- **Descending thin limb:** permeable (**AQP1**) — ~10–15%; water pulled out into the hyperosmotic medulla (**NO SOLUTE REABSORPTION HERE)**
- **Thick Ascending Limb / early DCT: WATER-IMPERMEABLE** → *"diluting segment"* (NaCl leaves, water stays)
- **Collecting Duct:** permeability **CONTROLLED BY ADH** (via **AQP2**) —*key hormone of water balance*

3) Water moves down its osmotic gradient into the interstitium — unlike Na^+, it is NEVER actively pumped

4) Water enters nearby capillaries (peritubular in cortex, vasa recta in medulla) → reclaimed into bloodstream



Tubular lumen
↓ **(Permeable wall; AQP1 / ADH-gated AQP2)**
Tubular epithelial cell
↓ **(via osmosis, basolateral AQP3/4)**
Renal interstitium
↓
Peritubular capillary / vasa recta
↓
Bloodstream

Medullary Gradient & Water Handling

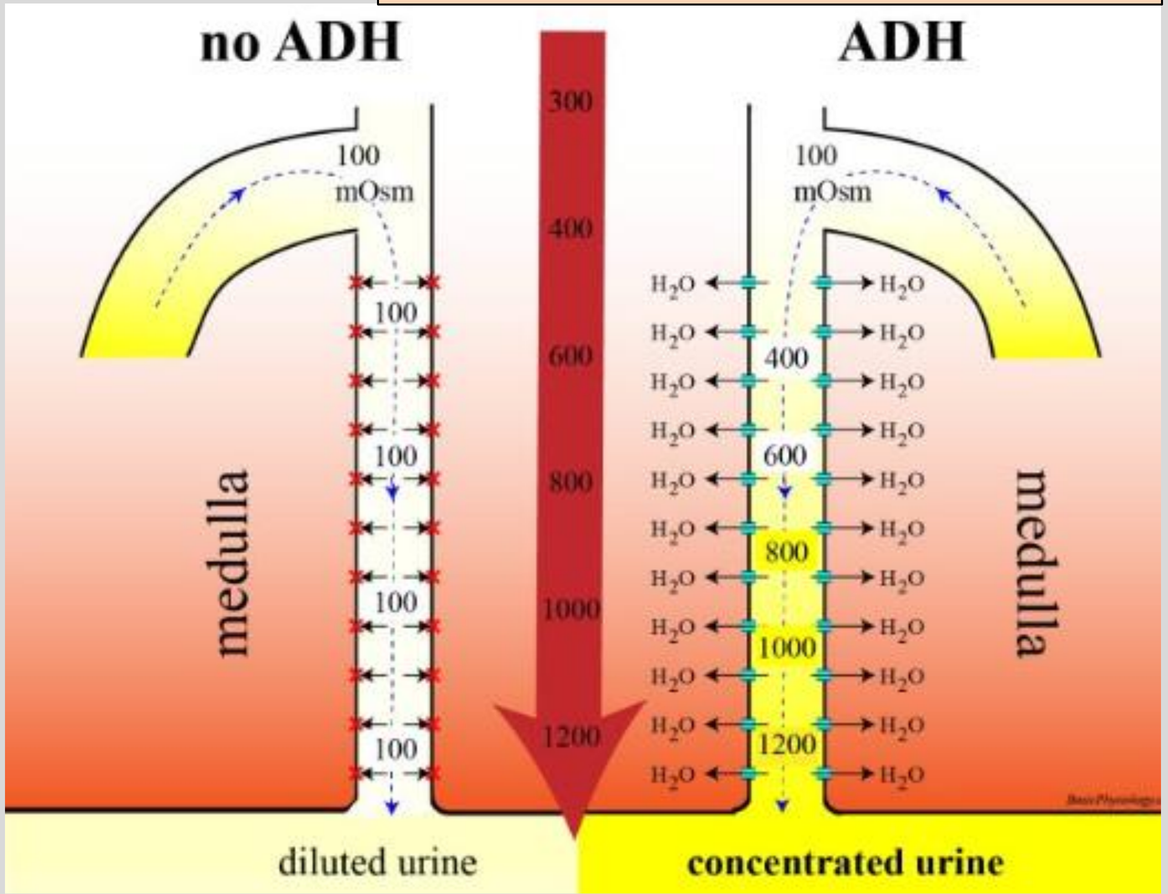
More on this in the urine concentration & dilution lecture…

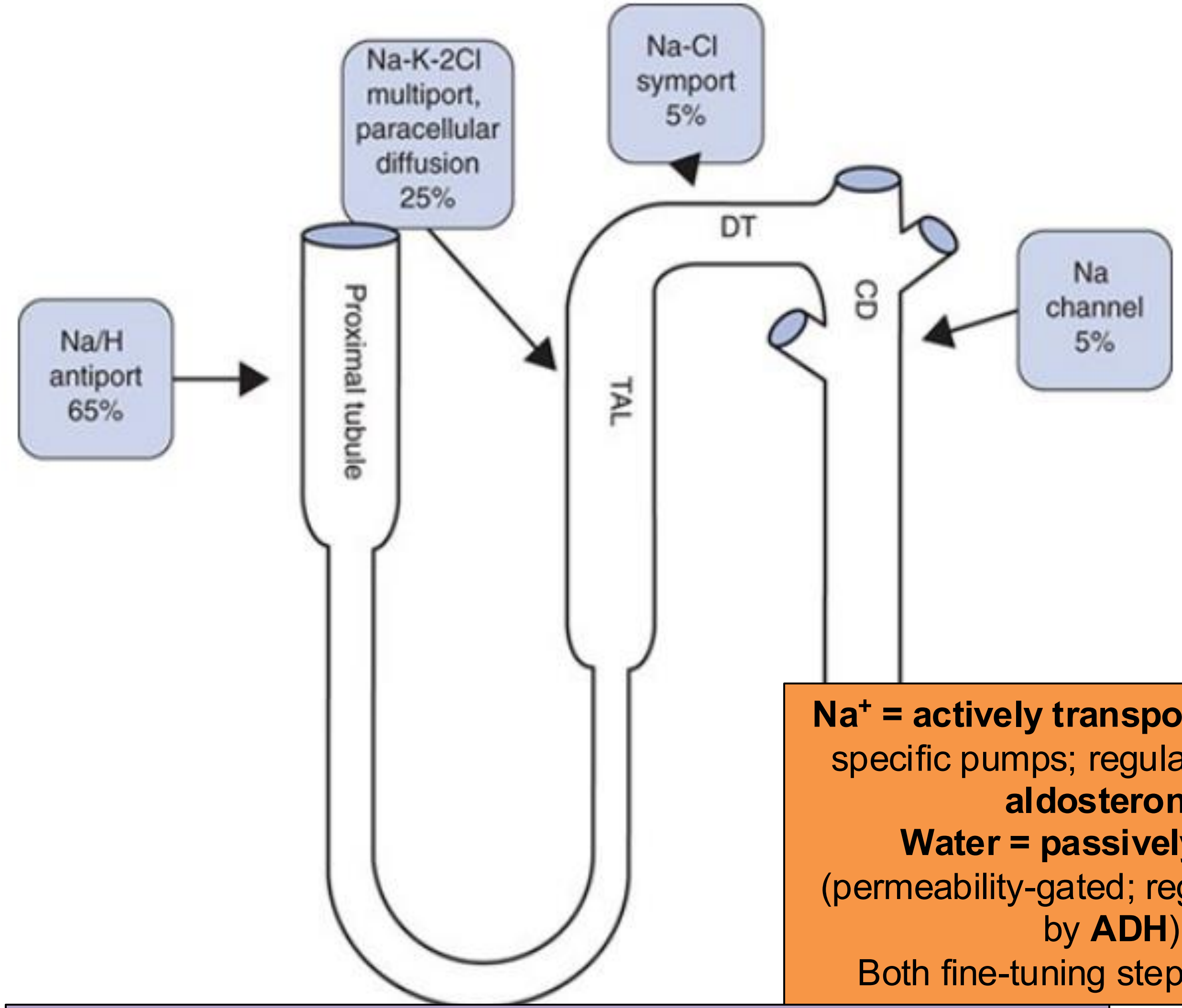
See Vander's Renal Physiology, Ch6, pg 14-19

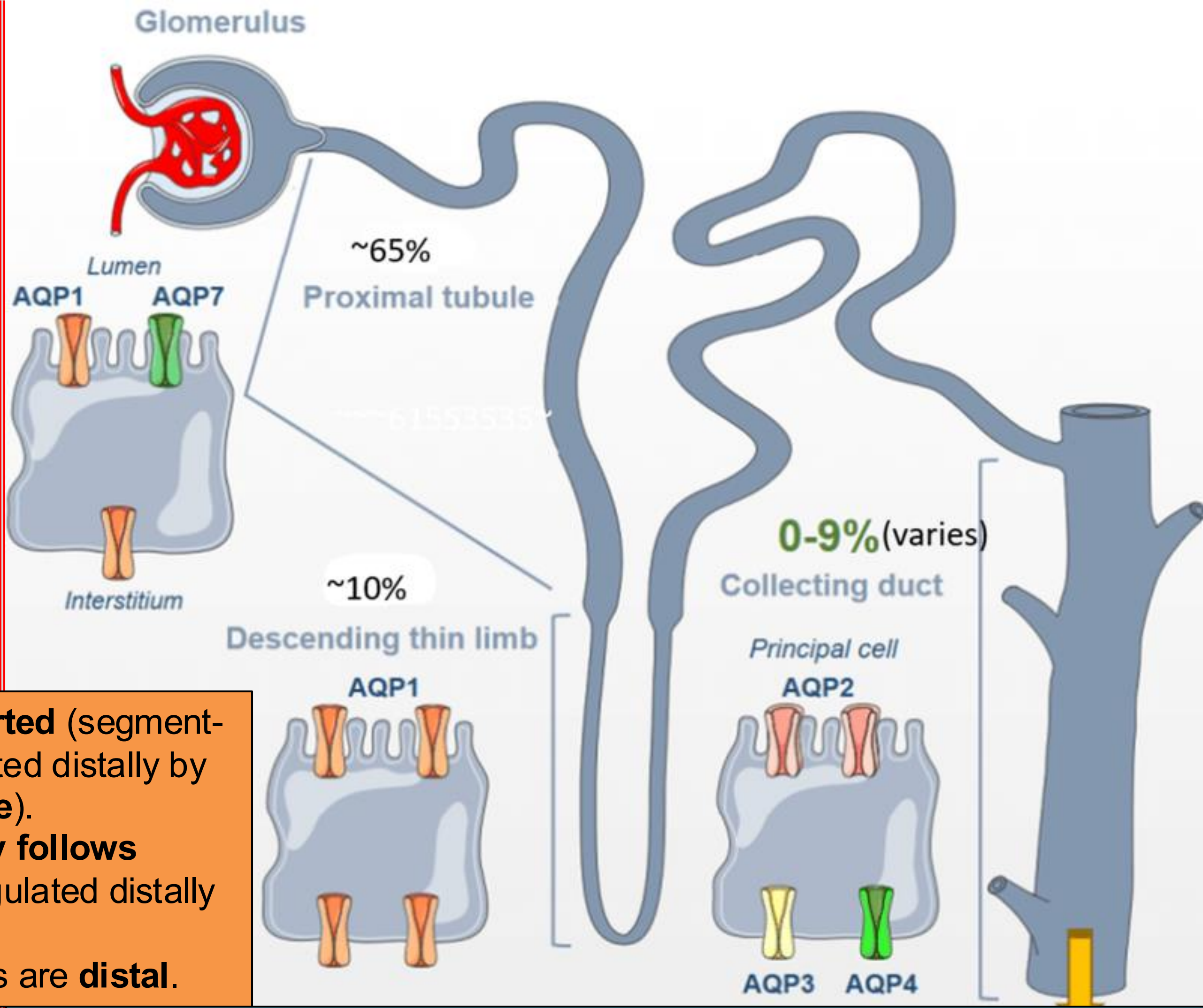
<u>Countercurrent Multiplication</u>:

Loop of Henle separates salt from water

- **Water impermeable TAL** pumps NaCl into renal medulla → hyperosmotic medullary gradient established
- At the collecting duct: ADH decides whether gradient is used (if water gets reabsorbed):
  - ADH (V2 receptor) aquaporin-2 insertion (tubule water permeable) → water flows down gradient into interstitium → urine becomes concentrated
  - No ADH → tubule stays impermeable → urine becomes dilute
- This is what allows the kidney to control water excretion independently of sodium
- This is also what makes loop diuretics the most powerful diuretic (collapses the gradient by blocking NKCC at TAL)







Na^+ = actively transported (segment-specific pumps; regulated distally by **aldosterone**).
Water = passively follows (permeability-gated; regulated distally by **ADH**).
Both fine-tuning steps are **distal**.

Tubular lumen
↓ (apical membrane)
Tubular epithelial cell
↓ (basolateral Na-K-ATPase)
Renal interstitium → Peritubular capillary / vasa recta → Bloodstream

Tubular lumen
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Tubular epithelial cell
↓ (via osmosis, basolateral AQP3/4)
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Nephron segment	Sodium reabsorbed	Water reabsorbed	Apical Transporter(s) Used	Main ideas / Clinical Applications
Proximal Convoluted Tubule	~65%	~65%	Sodium: **NHE3, SGLT2, Na-aa** Water-permeable: **AQP1** channel	**Isotonic bulk reabsorption**; the volume of the luminal fluid decreases, but osmolarity stays constant (water and sodium leave together) Drugs: Acetazolamide (indirect), SGLT2 inhibitors (diabetes)
Descending thin limb	**None**	~10% water	Water-permeable: **AQP1**	Water leaves; tubular fluid concentrates
Ascending limb (TAL)	~25% NaCl	**None**	**NKCC2** Water-impermeable	"Diluting segment" → NaCl reabsorbed but not water Loop diuretic segment CC: Bartter syndrome
Distal Convoluted Tubule	~5% NaCl	**None (early DCT)**	**NCC**	Thiazide-sensitive segment (acts on NCC) CC: Gitelman syndrome
Collecting duct	~4–5% NaCl	Variable	**ENaC (Aldosterone)** Water: **AQP2 (ADH)**	**Fine-tuning** by aldosterone (Na+) and ADH (water) Drugs (K+ sparing diuretics): spironolactone / eplerenone (aldosterone antagonists), amiloride / triamterene (ENaC blockers) CC: Liddle Syndrome

Big Picture - The Kidney and Cardiovascular System = An Important Partnership

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<u>The Kidney and Cardiovascular System Depend on Each Other to Maintain Blood Circulation</u>:

The kidney filters blood; to do so, the <u>cardiovascular system must provide enough blood pressure</u> to push plasma into the glomerulus

- Heart pumps blood → arterial pressure generated → blood reaches renal arteries → pressure in glomerular capillaries pushes fluid into Bowman's space → GFR occurs

The kidneys <u>maintain blood volume (in the long term)</u>, mainly by deciding how much <u>sodium and water</u> to keep or excrete

- More sodium retained → water follows → ECF volume increases → plasma volume increases → vascular system is better filled
- More sodium excreted → water follows → ECF volume decreases → plasma volume decreases → vascular filling decreases
- Kidney also secretes erythropoietin (EPO) which stimulates production of RBCs (~45% of blood volume)

Cardiovascular Primer for Renal Regulation

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Effective blood circulation for organ perfusion requires:

1) Enough **pressure** to drive blood forward
2) Enough **blood volume** to fill vascular system (ECF volume)

<u>1) Pressure</u> depends on **flow** (how much blood pumps forward, or CO) and **resistance** (of the blood vessels when blood is pumped, or TPR)

- This is conceptualized clinically by **Mean Arterial Pressure** = **Cardiac Output (CO) x Total Peripheral Resistance (TPR)**
  - In words, MAP describes the average pressure pushing blood through your arteries to perfuse vital organs
    - (the BP we measure with a sphygmomanometer measures the force against artery walls)
  - Normal = 70-100mmHg
- **Flow = Cardiac Output = Heart Rate** (**HR**, in bpm), **x Stroke Volume** (**SV**; blood volume pumped per beat)
- **Resistance = Total Peripheral Resistance** → determined by **constriction or relaxation of systemic arterioles**
  - Systemic Arterioles = major vessel of resistance;
    - When they constrict → vessel radius smaller → inc. resistance → inc. pressure
    - When they relax/dilate → vessel radius larger → dec. resistance → dec. pressure

Effective blood circulation for organ perfusion requires:
1) Enough **pressure** to drive blood forward
2) Enough **blood volume** to fill vascular system (ECF volume)

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2) Blood Volume (plasma volume within the ECF) depends on the factors influencing **sodium and water balance**

→ **Renal and renally-related hormonal sodium/water balance is main long-term determinant of this volume**, but several other factors influence, such as:
- → Fluid intake/restriction
- → Fluid loss outside kidney: sweating, diarrhea, vomiting, bleeding, insensible loss through skin/lungs
- → FLUID SHIFTS ** (think starling forces; i.e. if there is low albumin → dec. plasma oncotic pressure → fluid leaves capillaries and shifts into interstitial space → edema with reduced effective circulating volume)
- → RBC mass (the other 55% of blood volume)

→ The heart can only pump what returns to it = the venous return

→ This ultimately decides stroke volume, or the volume of blood pumped per heartbeat

MAP = CO x TPR
CO = HR x SV

Effective blood circulation for organ perfusion requires:

1) Enough **pressure (MAP)** to drive blood forward
   - Determined by **flow (Cardiac Output)** and **Resistance (Constriction/Relaxation of the Arterioles** of which blood is pumping towards)
2) Enough **blood volume** to fill vascular system (ECF volume)

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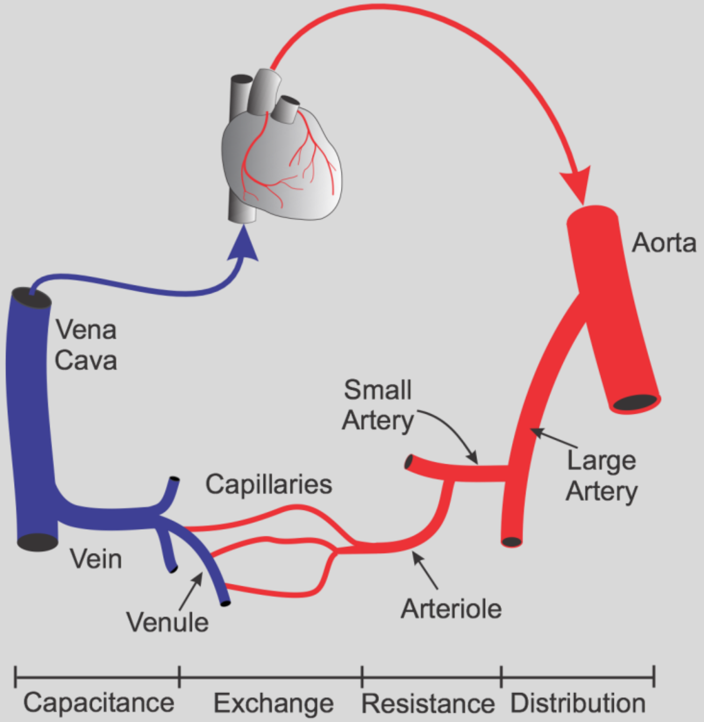
- Renal Na+ / Water Balance
- Nonrenal Fluid Loss
- Plasma / Interstitial Shifts
- Fluid intake
- RBC mass

Blood Volume

Venous Return (blood returning to heart)

Preload



Stroke Volume

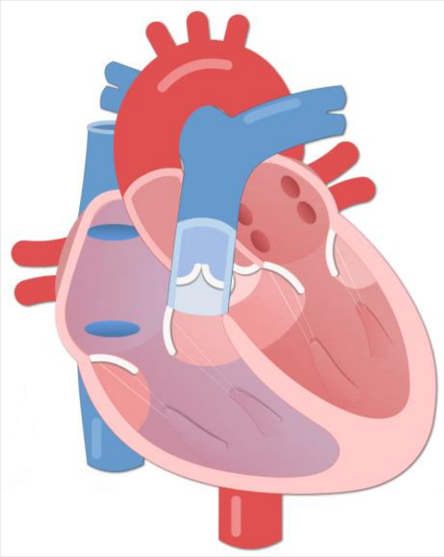
Cardiac Output

Arteriole Tone

Total Peripheral Resistance

Blood Pressure (MAP)

(Preload = amount of ventricular filling before contraction)



MAP = CO x TPR
CO = HR x SV

Effective Circulating Volume (ECV)

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- Effective Circulating Volume (ECV) = portion of arterial circulation **physically able to perfuse tissues** and being sensed by body's volume/pressure receptors
    - How much of the circulating volume is "effective"
- ECV $\neq$ Total ECF (Extracellular Fluid Volume)
- This concept will come in handy later when discussing pathological conditions later such as heart failure, where ECF is high (volume overloaded), but ECV is low (poor organ perfusion)

ECF = how much water is in the plumbing
ECV = how much water can reach the taps

Putting it all together…

Q1. A 52-year-old man with a history of high dietary sodium intake presents with persistently elevated blood pressure. **His physician explains that excess sodium can contribute to hypertension by increasing extracellular fluid volume**.

Which sequence best explains how increased total body sodium can raise blood pressure?

A. Increased total body sodium → increased ECF volume → increased plasma volume → decreased venous return → decreased preload → decreased stroke volume → decreased cardiac output

B. Increased total body sodium → increased ECF volume → decreased plasma volume → decreased venous return → decreased preload → decreased stroke volume → decreased blood pressure

C. Increased total body sodium → decreased ECF volume → decreased plasma volume → decreased venous return → decreased preload → decreased stroke volume → decreased cardiac output

D. Increased total body sodium → increased ECF volume → increased plasma volume → increased venous return → increased preload → increased stroke volume → increased cardiac output

E. Increased total body sodium → increased ECF volume → increased plasma volume → increased venous return → decreased preload → decreased stroke volume → decreased cardiac output

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Lecture Format

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Part 1: Where sodium and water move & why it matters

Part 2: How the body regulates those movements

Part 3: Why dysregulation causes disease

Overview:

1) Sensing → Baroreceptors, osmoreceptors, macula densa
2) Responding → RAAS, ADH, natriuretic peptides
3) Protecting Stabilizers → Autoregulation, TGF (Tubuloglomerular feedback), GTB (Glomerulotubular balance)

Goals of Regulation

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Kidney filters ~25,200 mmol of sodium a day (≈1.5 kg of salt)
Adult salt intake = ~100–250 mmol
Kidney must reabsorb >99% sodium and excrete a tiny, chosen remainder.

- Being off by the smallest % is a crisis: reabsorbing 1% less dumps liters of ECF → shock; 1% more → edema and hypertension

Kidney's Goal: **Reabsorb 99% of Na^+ reliably, then fine tune the last 1% (to adjust blood volume, blood pressure)**

Goal #1 → Keep GFR steady

Goal #2 → Keep Na^+ delivery to distal nephron consistent

Goal #3 → Vary distal Na^+ reabsorption to match output/intake (per Volume/Osmolality signals) (via RAAS/aldosterone/ADH)

By keeping the proximal nephron's **Na^+** conditions steady, the distal nephron can be the adjustable valve to match output and intake

Three Timescales of BP Control:

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Timescale	Mechanism	Adjusts:
Short (sec – min)	**Arterial baroreflex** (autonomic reflex)	Heart rate, contractility, vascular tone
Intermediate (min – hr)	Angiotensin II actions of **RAAS** (Renin-Angiotensin-Aldosterone System)	Vascular Resistance (chemical vasoconstriction)
Long (hr – days)	**Renal Na^+ & Water Handling** (adjusting distal sodium reabsorption) Actions of Aldosterone	Actual ECF / Blood Volume

Universal goal: Defend MAP (keep just under 100mmHg)

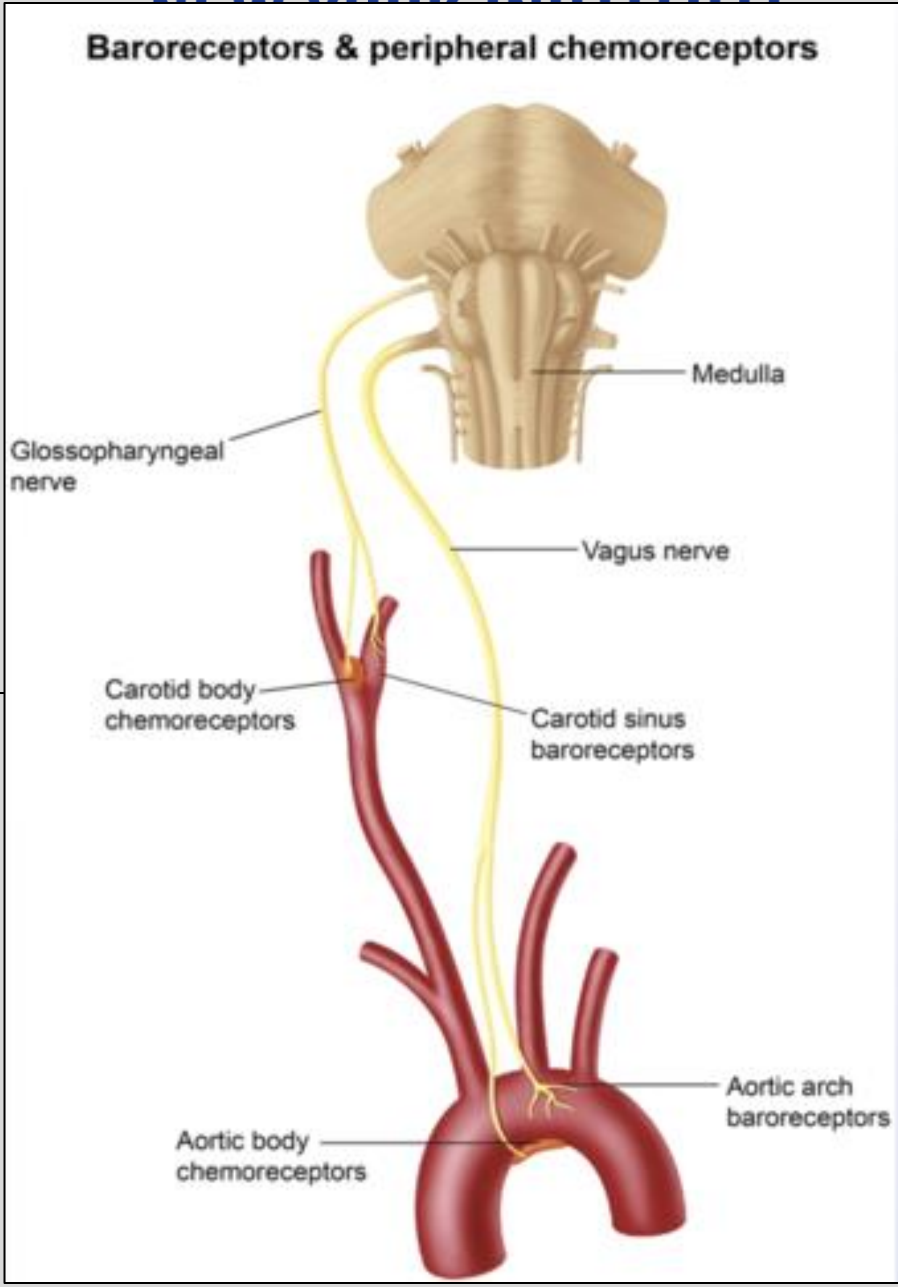
MAP = CO x TPR
CO = HR x SV

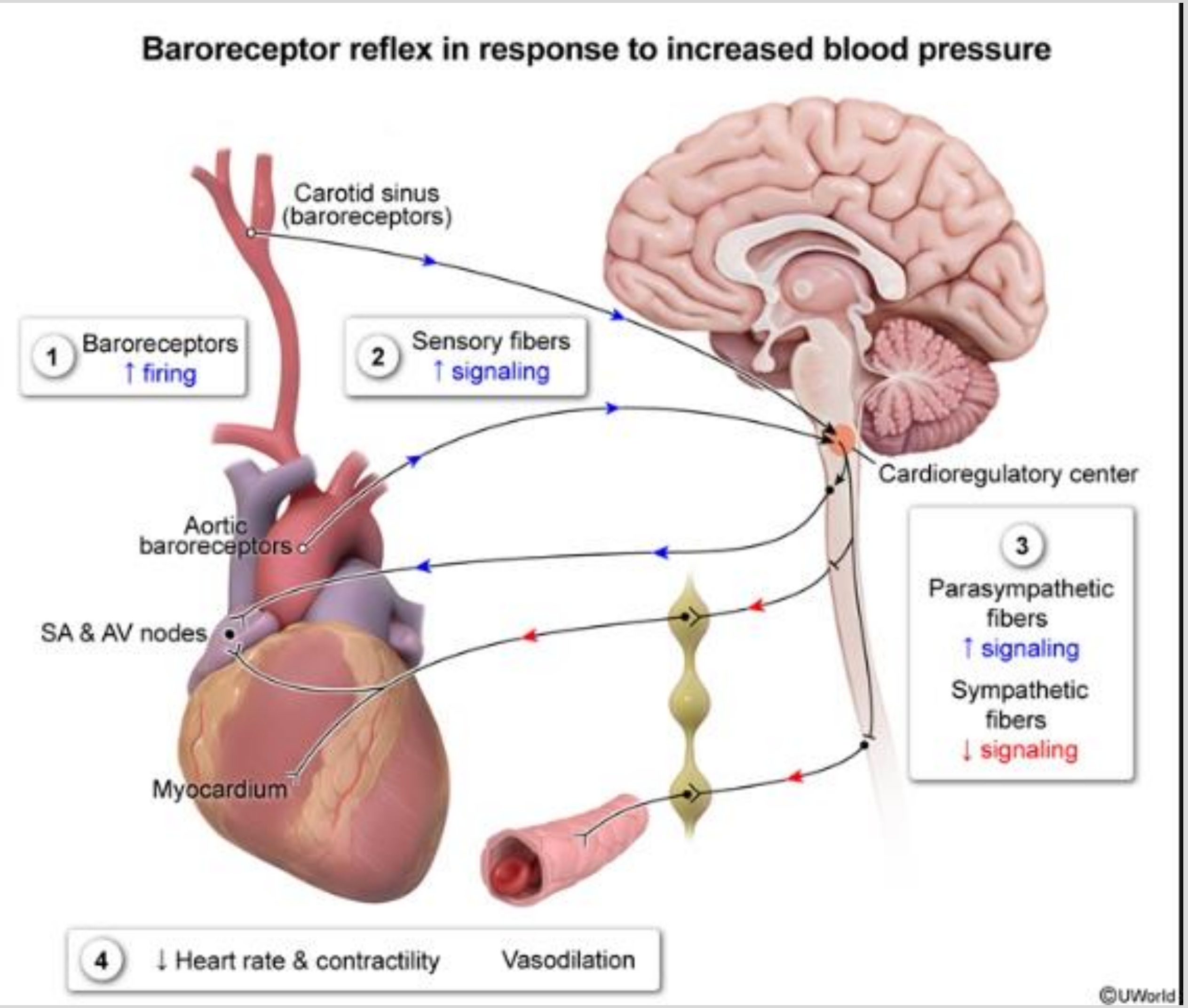
Arterial Baroreceptors – Short Term BP Control

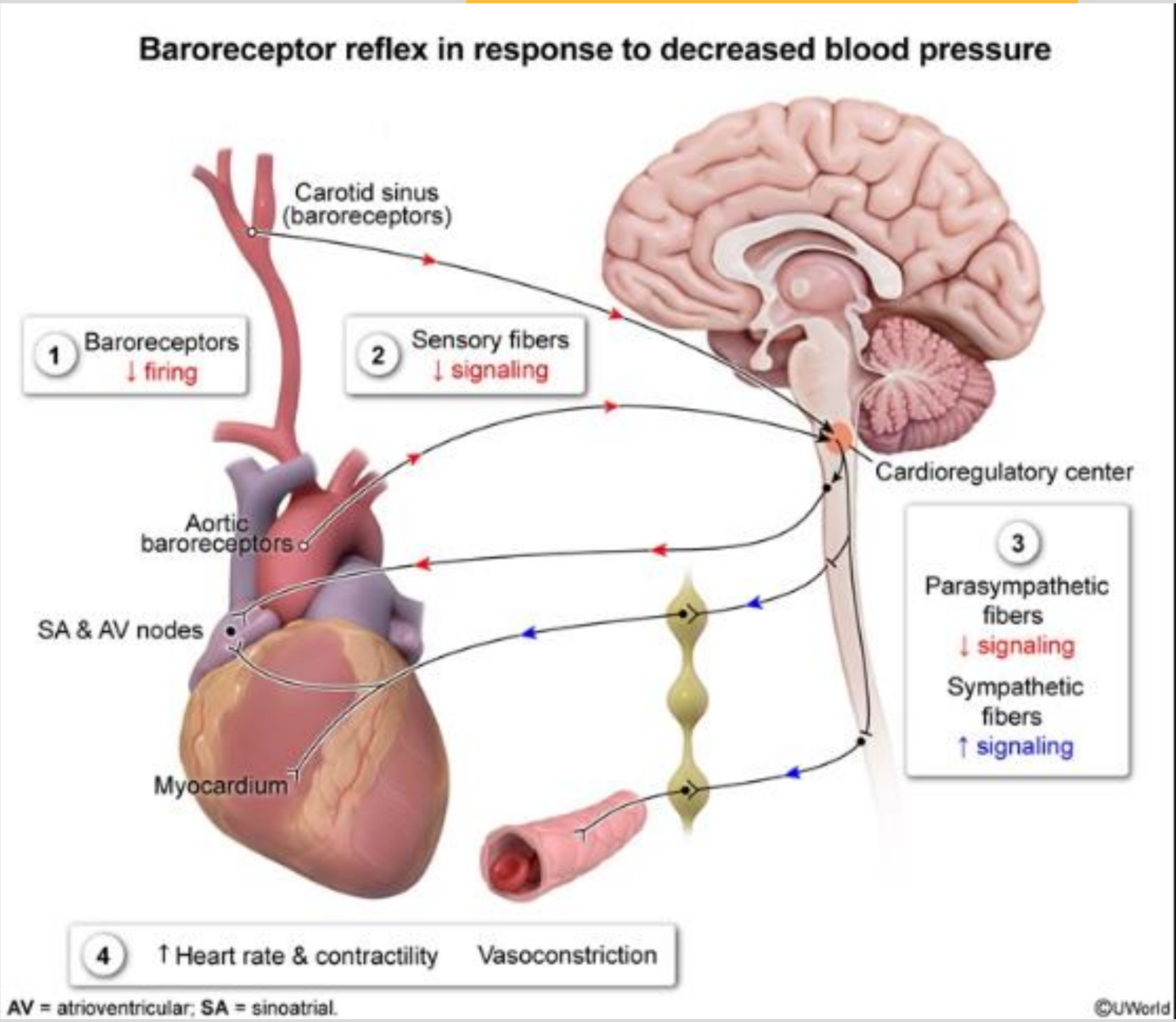
- Stretch-sensitive mechanoreceptors that **increase** their firing rate in response to **increased arterial distention** (↑ BP)
- **Carotid Sinus** (just above carotid bifurcation)
  - Afferent signals carried via Glossopharyngeal nerve (CN IX)
- **Aortic Arch**
  - Afferent signals carried via Vagus nerve (CN X)
- Signals sent to **medulla** to modulate sympathetic and parasympathetic (vagal) output:

- ↑BP → ↑arterial wall stretch → ↑baroreceptor firing → ↓sympathetic tone, ↑parasympathetic outflow
  - Vasodilation, decreased heart rate and contractility → BP falls

- ↓ BP → ↓ arterial wall stretch → ↓ baroreceptor firing → ↑ sympathetic tone, ↓ parasympathetic outflow
  - Vasoconstriction, increased heart rate and heart contractility → BP rises







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- Vasodilation, decreased heart rate and contractility → BP falls

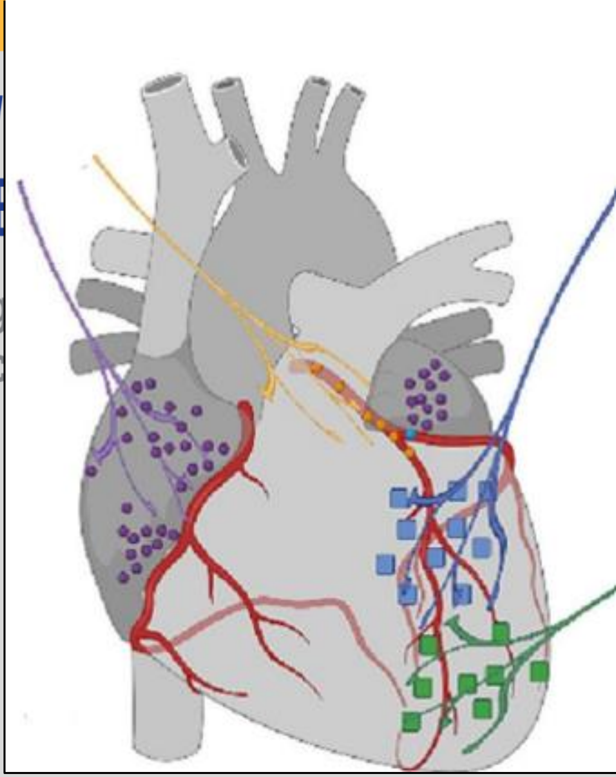
↓ BP → ↓ arterial wall stretch → ↓ baroreceptor firing → ↑ sympathetic tone, ↓ parasympathetic outflow

- Vasoconstriction, increased heart rate and heart contractility → BP rises

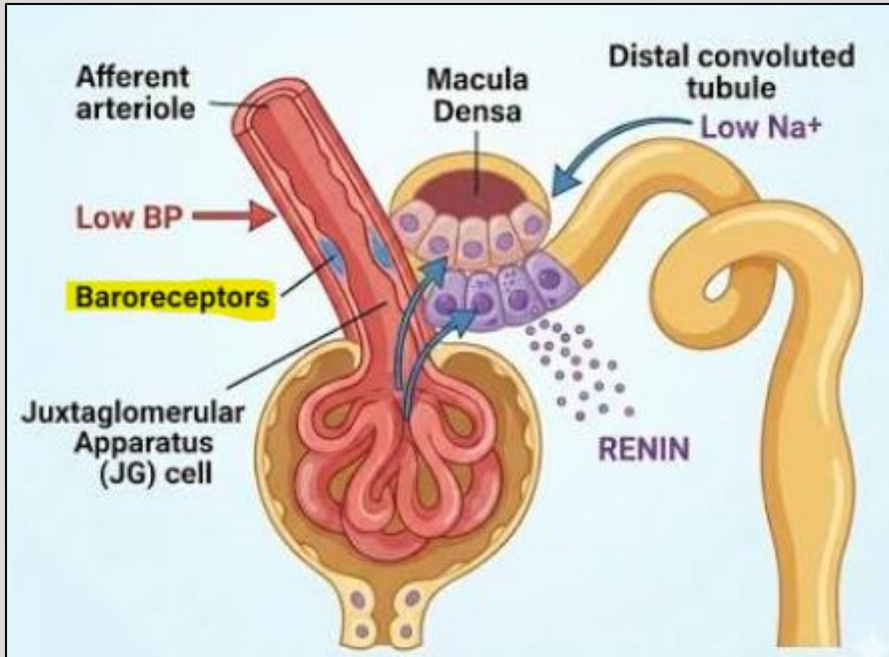
Other Vascular Baroreceptors

Sensory signals from these baroreceptors feed the slower hormonal systems:

- Cardiopulmonary baroreceptors (on atria, ventricles, pulmonary vasculature): detect venous filling (neural volume sensors) → strongly influence ADH, sympathetic tone



- Intrarenal baroreceptors = **Juxtaglomerular (JG) (granular)** cells in **renal afferent arteriole**: low perceived afferent pressure influences release of renin and **activates RAAS** pathway (next slides)



The RAAS (Renin-Angiotensin Aldosterone System: Overview

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RAAS switches **on** when body senses it is **under-filled or under-perfused**; defends arterial blood pressure and effective circulating volume by

a) Retaining Na^+ and water
b) Constricting vessels

<u>The cast, in sequence (*note where each is made*):</u>

->Angiotensinogen — precursor protein from the **liver**; always in excess (not rate-limiting).

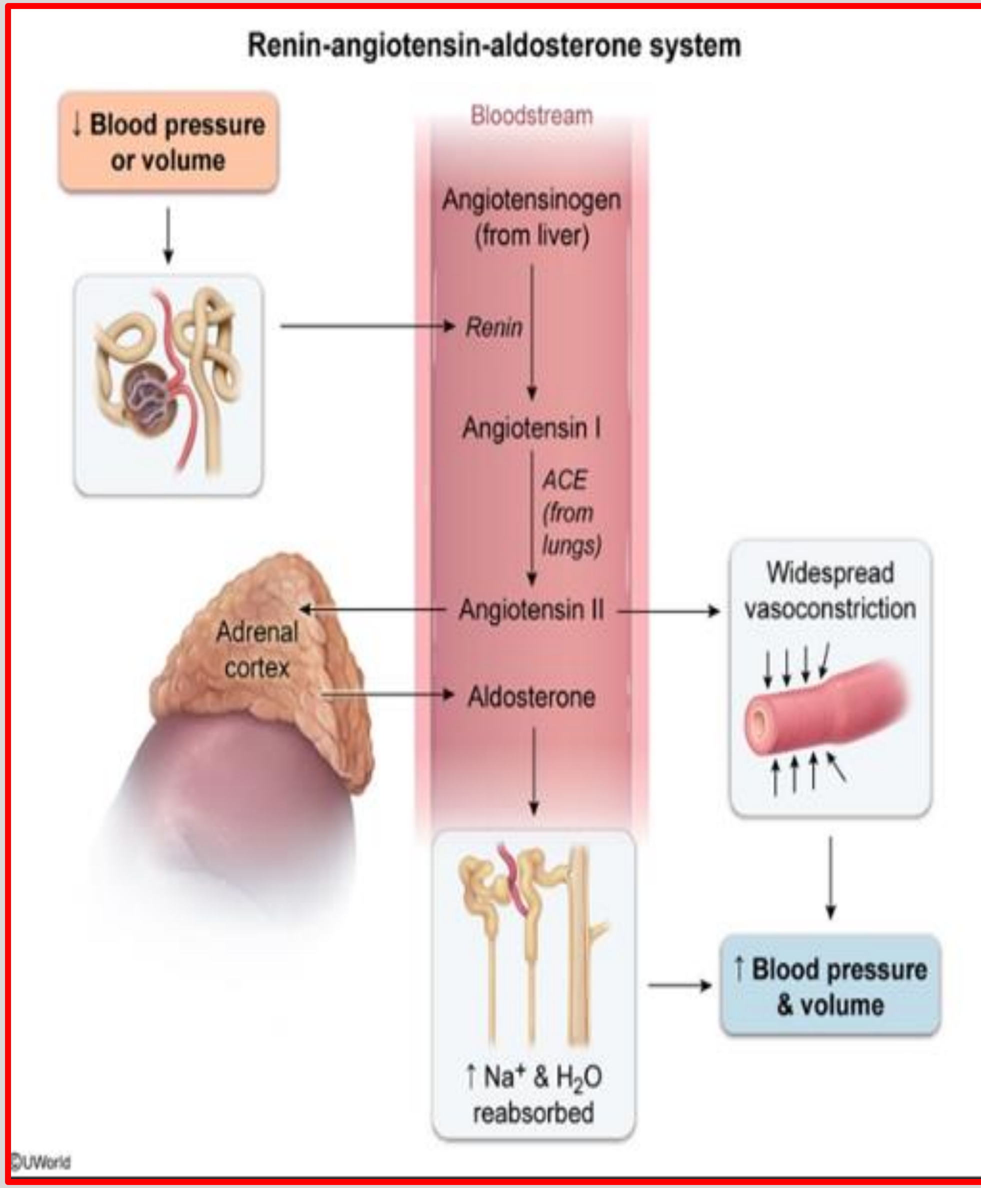
->Renin — enzyme from the **juxtaglomerular (JG) cells** of the afferent arteriole; the **rate-limiting step**

->Angiotensin I — inactive intermediate (a precursor).

->ACE (angiotensin-converting enzyme) — mainly on **endothelium of the vessels in the lung**.

->Angiotensin II — the active effector hormone

->Aldosterone — steroid hormone from the **adrenal cortex**

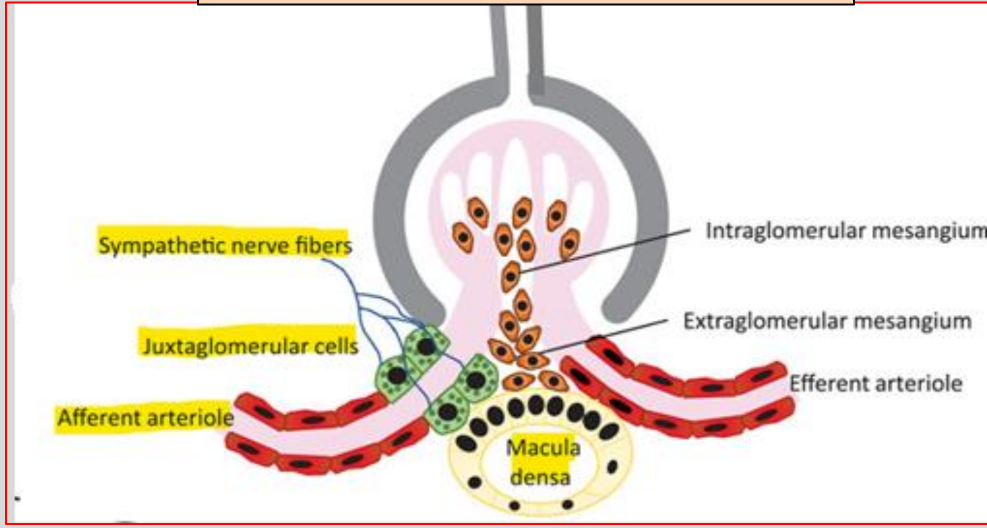


RAAS Walkthrough (Part 1): Renin + Conversion

1) **Juxtaglomerular apparatus (JGA)** causes renin release in response to 3 triggers:

- **Renal sympathetic nerves: β1** receptors on JG cells → ↑renin.
- **Intrarenal baroreceptor:↓afferent arteriolar pressure → ↑renin.**
- **Macula densa: low** DCT NaCl delivery → prostaglandins → **↑renin**

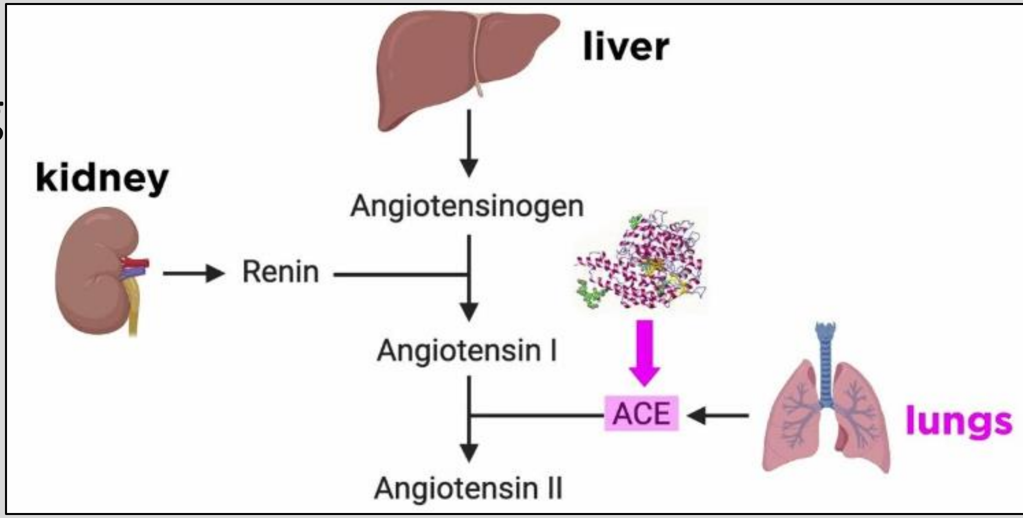
Juxtaglomerular Apparatus:



2) **Renin released** by JG (granular cells) of afferent arteriole

- Renin is the enzyme serving as the rate limiting step of this cascade

3) Renin **cleaves liver-derived angiotensinogen**, converting it to **Angiotensin I (Ang I)**



4) Ang I is converted to Angiotensin II (Ang II) by ACE (Angiotensin converting enzyme), located within the endothelium of **lung** vessels

RAAS Walkthrough (Part 2): Actions of Ang II

5) **Angiotensin II** is a **peptide hormone** binding to **AT1 receptors** (Gq-coupled GPCR) to perform the following actions:

- **↑ ↑ Vasoconstriction** of systemic arterioles: ↑TPR → ↑BP
- In Kidney: Constricts **efferent arteriole** > afferent arteriole
  - This allows GFR to be preserved even in low volume states (ex. hemorrhage)
- **↑ Reabsorption of Na^+** by stimulating **NHE3** (proximal), **NCC** (early distal), **ENaC** (late distal/collecting).
- Stimulates **aldosterone release** from the **adrenal cortex**
- Communicates to CNS to **↑ Thirst, Salt Appetite** **(hypothalamus), Sympathetic drive, ADH release (post. pituitary)**
  - **ADH release → ↑ AQP2 expression → ↑ water reabsorption at collecting duct**

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More AT1 receptors are on the efferent arteriole, which is why AngII has this preference!

Clinical Pearl: ACE-inhibitors (inhibiting ACE) and Angiotensin Receptor Blockers (ARBs) are a 1st line treatment in hypertension, heart failure, diabetic kidney disease. ARBs block AT1 directly.

(more in Pharmacology lectures)…

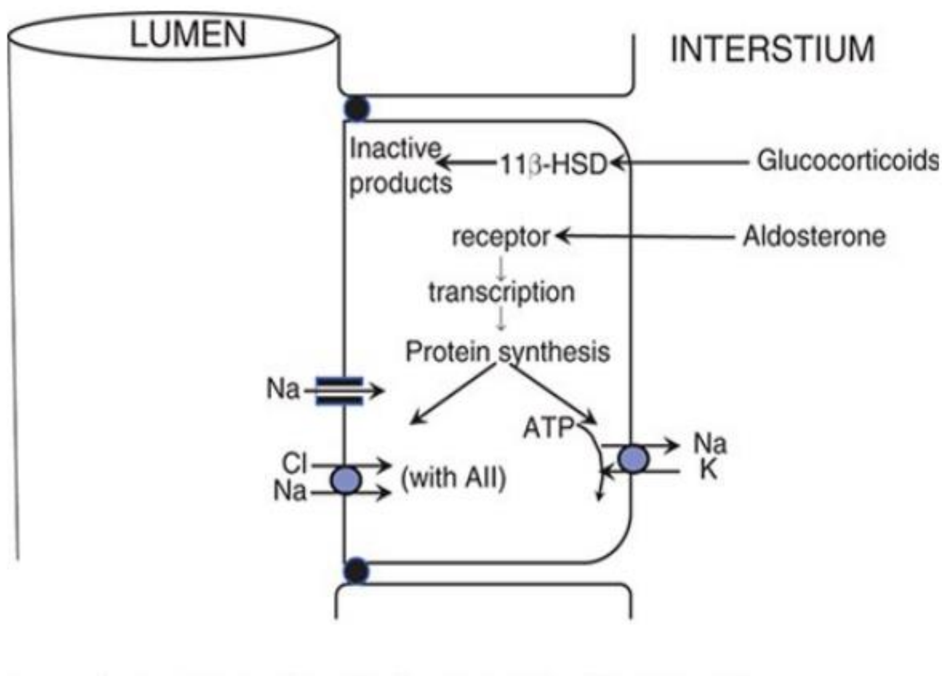
RAAS Walkthrough (Part 3): Actions of Aldosterone

6) **Aldosterone** (activated by Ang II) is a **steroid hormone** that binds to intracellular **mineralocorticoid receptor (MR)** → inducing transcriptional changes causing the following:

- ↑**ENaC** + ↑basolateral **Na-K-ATPase** in distal-nephron **principal cells**
  - **Causes ↑ Na^+ reabsorption** (water follows → ECF volume ↑) with **↑ K^+ excretion** (potassium wasting)
  - Also acts on colon, sweat, and salivary ducts to retain Na^+
- **When Ang II present, also** ↑NCC in DCT
- Also acts on α-intercalated cells to **↑ H^+ secretion** (see acid-base lecture)
- **Glucocorticoids (cortisol) can also bind to MR receptor;** however, they are inactivated by 11β-HSD, **protecting the MR** for aldosterone
  - Chronic licorice consumption inhibits 11β-HSD → MR receptor activated by cortisol → apparent mineralocorticoid excess (hypertension, hypokalemia, low renin, low aldosterone)

Recall that **peptide hormones (Ang II) have a **much shorter duration of effect** compared to **steroid hormones** (Aldosterone)

Diuretics sparing K^+:
Spironolactone and Eplerenone = MR antagonists
Amiloride, Triamterene = ENaC blockers
(more in Pharmacology lectures)…



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

RAAS Walkthrough (Part 4): Negative Feedback

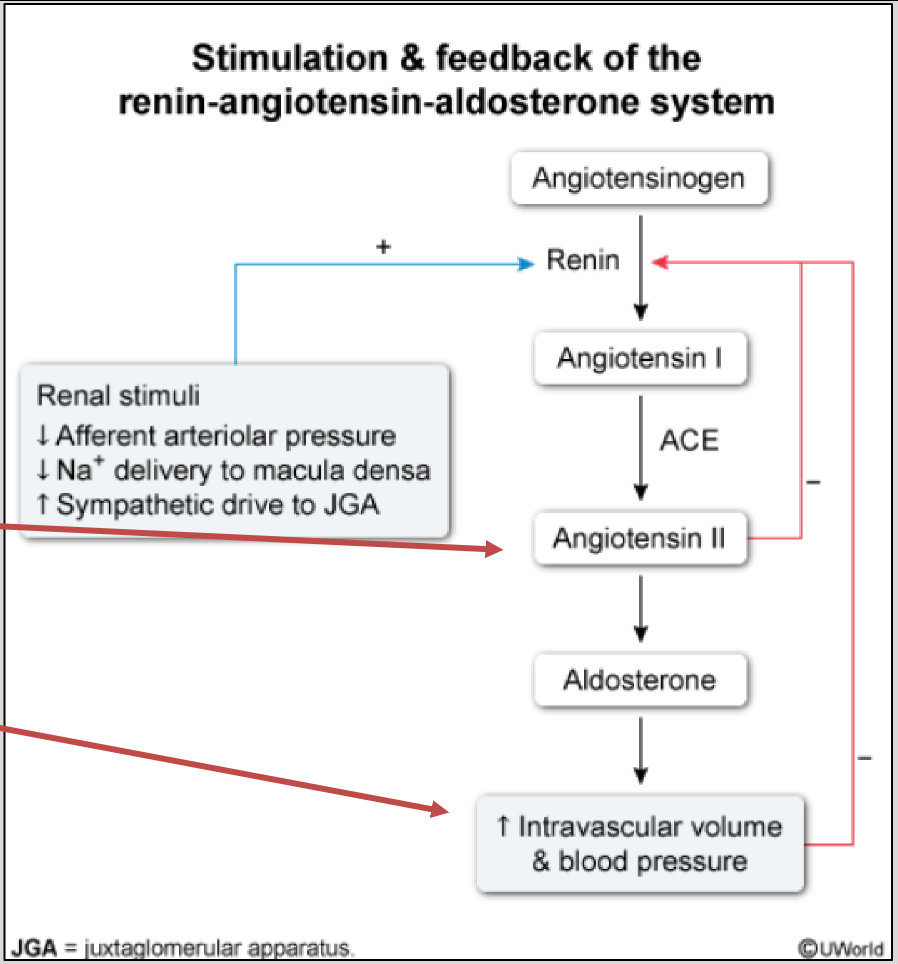
Recall the 3 renin-secreting stimuli (Slide 28)
↓ afferent arteriolar pressure
↓ Na^+ delivery to the macula densa
↑ sympathetic drive to the JGA (β1)

- Once RAAS restores effective intravascular volume/blood pressure, it utilizes a closed-loop negative feedback system to turn itself off:

- Short loop: Ang II directly inhibits renin release from JG cells (feedback on its own production)

- Long loop: ↑ **intravascular volume & blood pressure** reverses all three renin-secreting stimuli → ↓ renin

- Once volume/pressure is corrected, the triggers vanish and renin falls. **If the underlying deficit persists (e.g., heart failure with low *effective* volume), the loop keeps firing = Pathologic RAAS Activation**



Renal Artery Stenosis

- Narrowed renal artery → kidney "sees" low pressure (low ECV) → **chronic renin release → chronic hypertension**
- Presents as worsening hypertension, patients often on **≥3 antihypertensives without improvement in BP**
- Caused by plaque buildup, or fibromuscular dysplasia (a rare fibrous overgrowth of arterial walls, females aged 15-50)
- Usually unilateral, but bilateral possible

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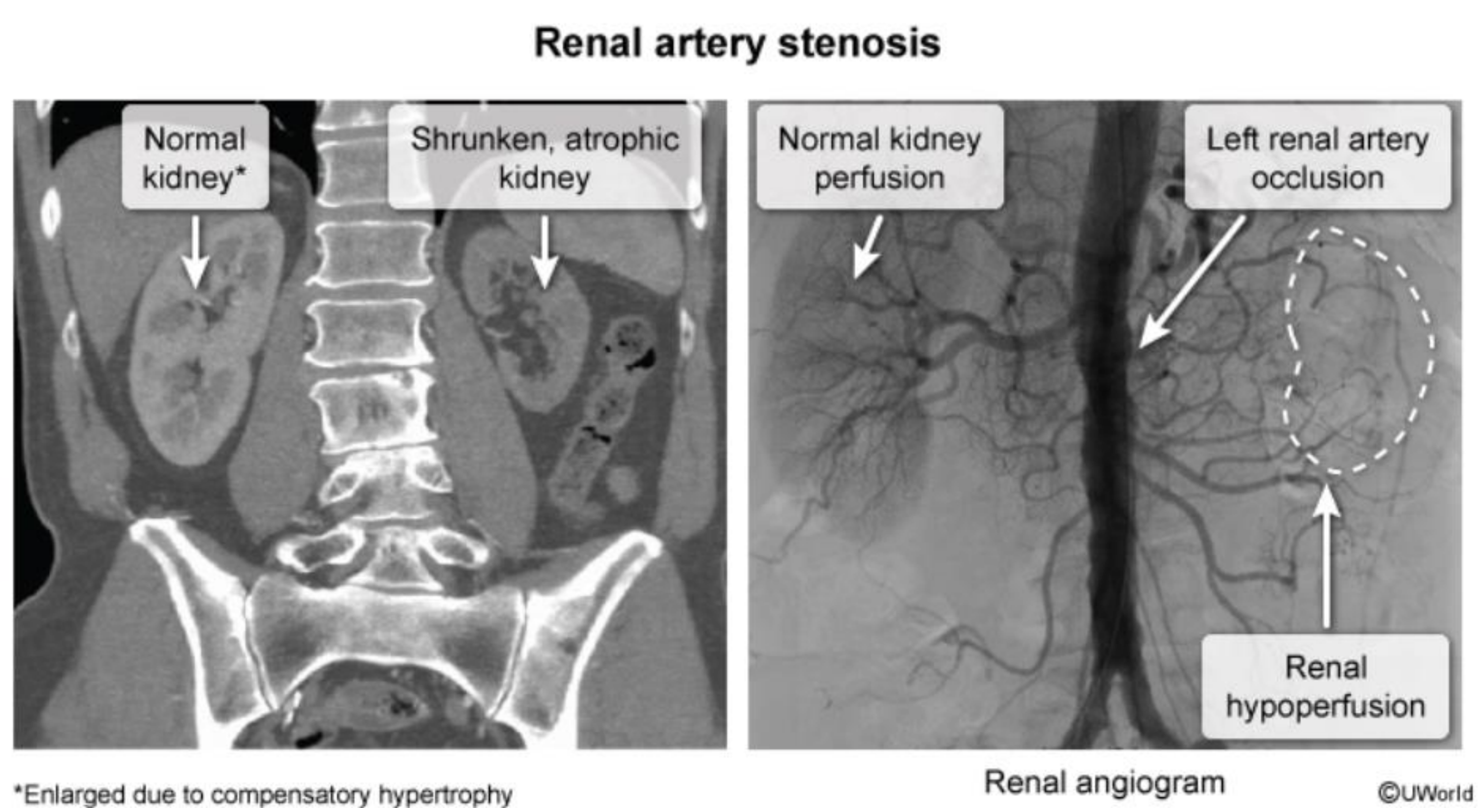
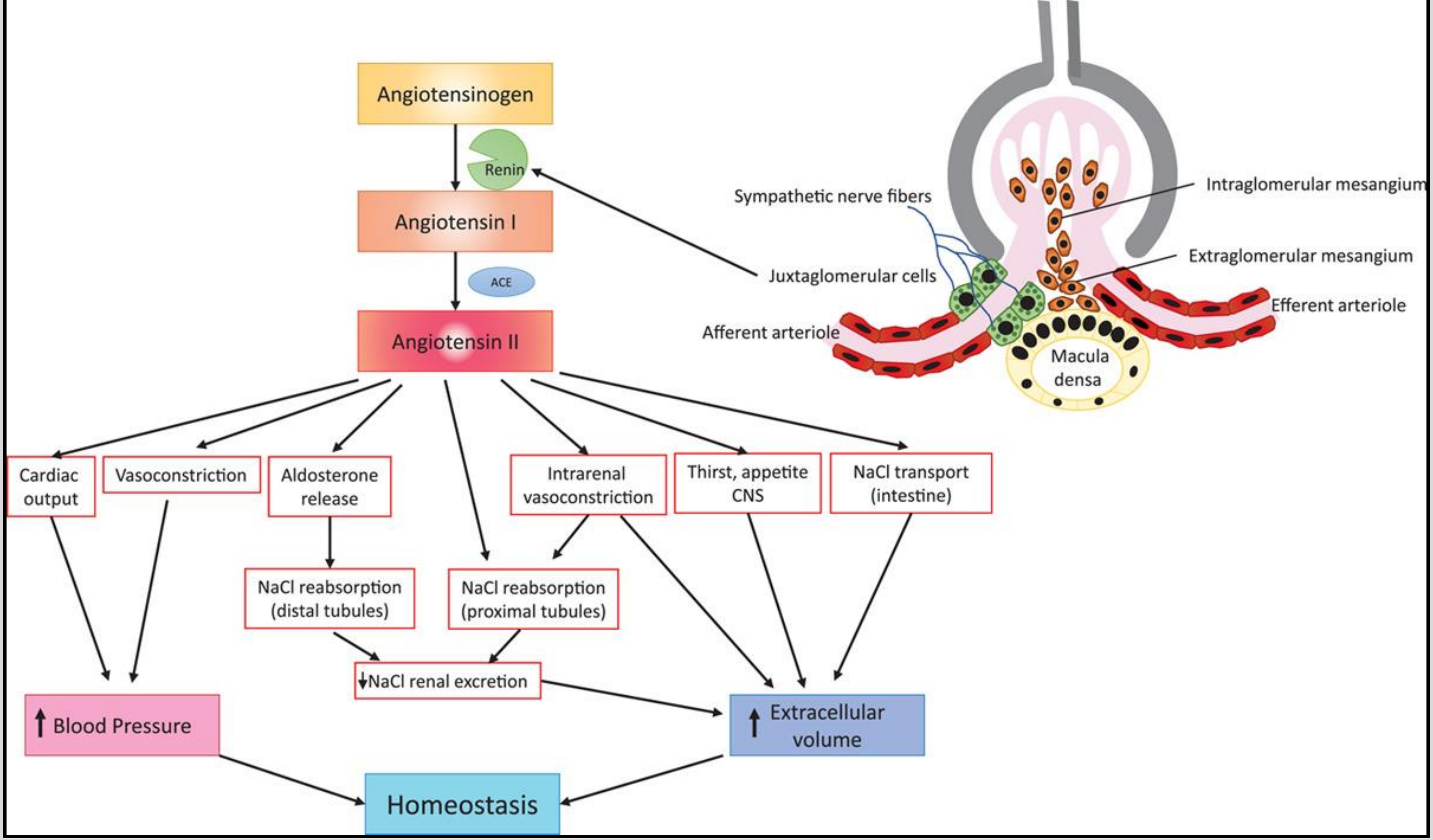



Angiotensinogen
Renin
Angiotensin I
ACE
Angiotensin II
Sympathetic nerve fibers
Intraglomerular mesangium
Juxtaglomerular cells
Extraglomerular mesangium
Efferent arteriole
Afferent arteriole
Macula densa
Cardiac output
Vasoconstriction
Aldosterone release
Intrarenal vasoconstriction
Thirst, appetite CNS
NaCl transport (intestine)
NaCl reabsorption (distal tubules)
NaCl reabsorption (proximal tubules)
↓NaCl renal excretion
↑ Blood Pressure
↑ Extracellular volume
Homeostasis

Factors Increasing Sodium Excretion (aka Natriuresis) - "The Anti-RAAS":

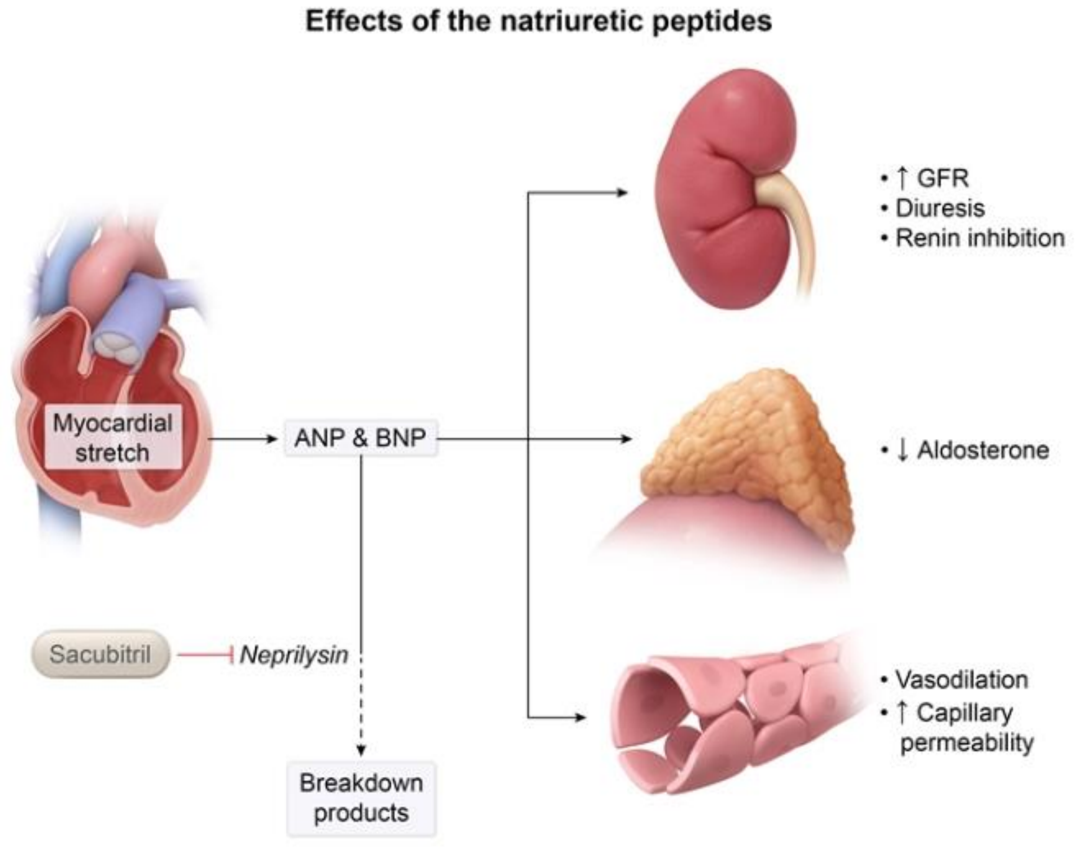
The Natriuretic Peptides: released from **cardiac myocytes** in response to ↑cardiac stretch/filling

- ANP (Atrial Natriuretic Peptide) – released from **atrial myocytes**
- BNP (Brain Natriuretic Peptide) – released from **ventricular myocytes**

Actions Oppose RAAS and Drive Na^+/Water loss:

- Dilate the **afferent arteriole** → ↑ GFR → ↑ filtered Na^+
- **Inhibit renin** and **inhibit aldosterone**.
- **Inhibit Ang II's** tubular Na^+ reabsorption
- Promote **vasodilation**

Net: ↑ GFR + ↓ reabsorption → **natriuresis + diuresis → ↓ ECF volume / BP**



ANP = atrial natriuretic peptide; BNP = brain natriuretic peptide; GFR = glomerular filtration rate.

Because BNP rises with ventricular stretch, it is a key biomarker in diagnosing heart failure

ANP/BNP broken down by enzyme Neprilysin; Sacubitril is a drug that is a neprilysin inhibitor, commonly paired with valsartan (ARB) to boost natriuresis while blocking RAAS in heart failure treatment

Additional Supporting Natriuretic Factors

- Intrarenal Dopamine
  - L-Dopa converted to dopamine in proximal tubule → acts as paracrine signaler to reduce sodium absorption via internalization of NHE3, Na-K-ATPase, and reducing AT1 receptor expression
- Endothelin
  - Produced by proximal tubular epithelial cells → inhibits NaCl reabsorption (mechanism unknown: may have to do with nitrous oxide (NO) production)
- Plasma [K^+] as a *switch*:
  - high K^+ → inhibits NCC (shifts Na^+ reabsorption to ENaC, favoring K^+ secretion)
  - low K^+ activates NCC

<u>The Stabilizers</u>: Protecting Goals #1 & #2

"3 Intrarenal Guardians"

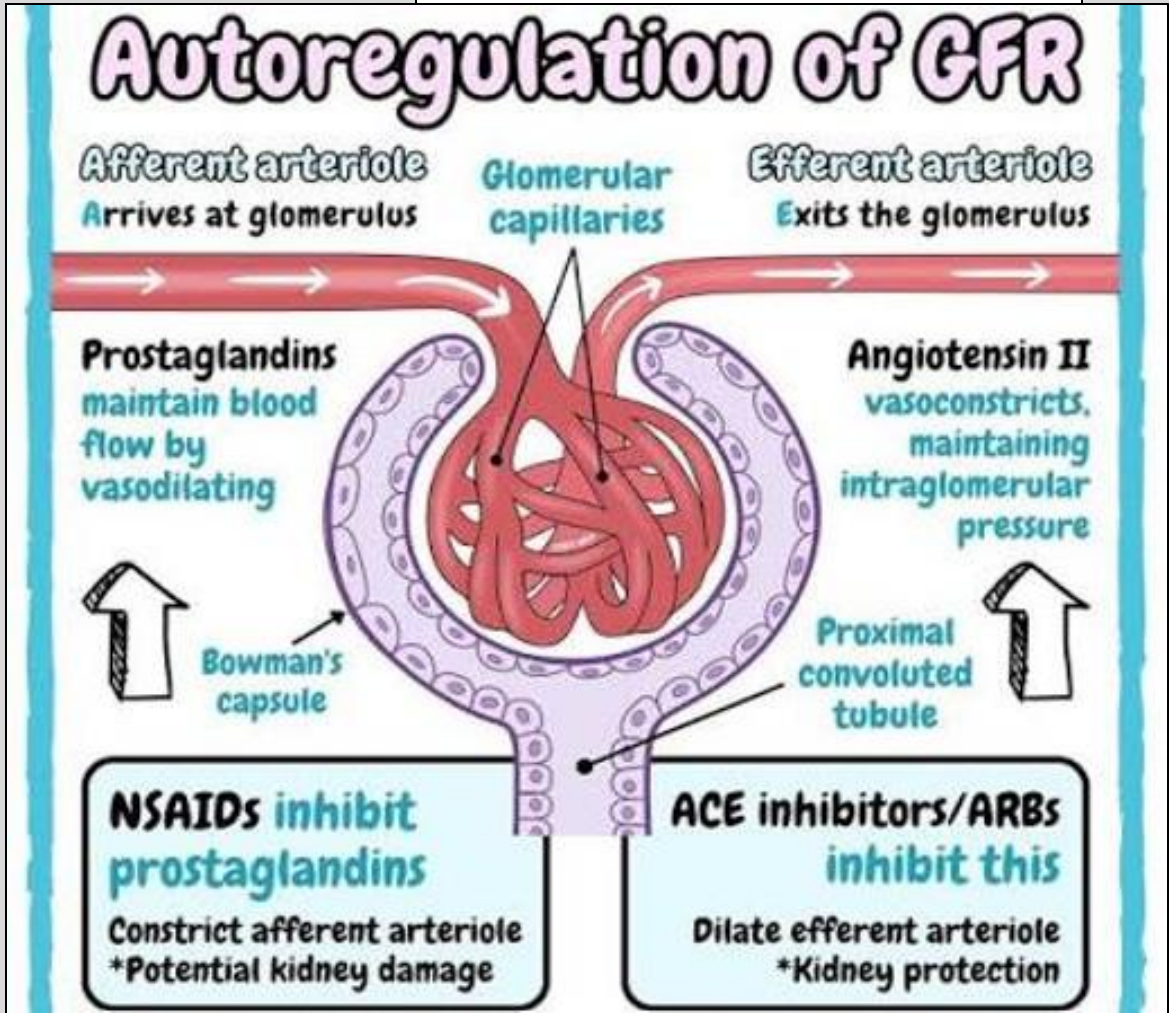
Recall Slide 20:

Goal #1 → Keep GFR steady
Goal #2 → Keep Na^+ delivery to distal nephron consistent
Goal #3 → Vary distal Na^+ reabsorption to match output/intake

1) Prostaglandins (PGE_2/PGI_2)

- Released by macula densa cells following Ang II stimulated vasoconstriction of renal arterioles (AngII vasoconstricts efferent > afferent arterioles)
- In response, PGE_2/PGI_2 selectively **dilates the afferent arteriole** to keep the inlet open and maintain RBF/GFR
- End result: Keeps RBF & GFR Steady (Goal #1)

- NSAID-Induced Nephrotoxicity
  - In volume-depleted patient, NSAIDs inhibiting COX and blocking synthesis of prostaglandins → unopposed afferent constriction → GFR crashes → AKI



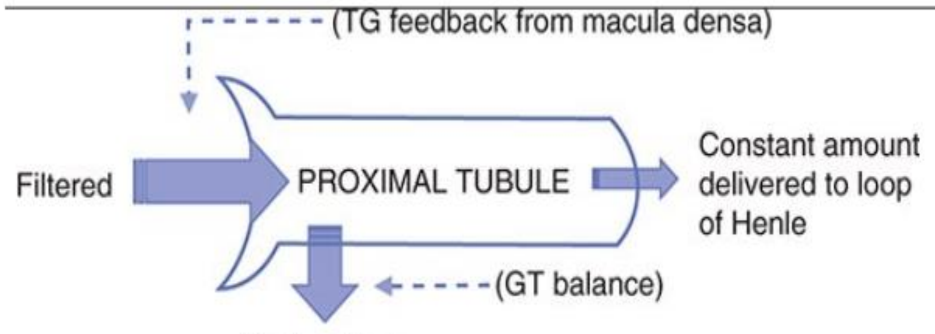
<u>The Stabilizers</u>: Protecting Goals #1 & #2 *"3 Intrarenal Guardians"*

2) **<u>Tubuloglomerular feedback (TGF)</u>** (Tubular feedback → glomerulus)

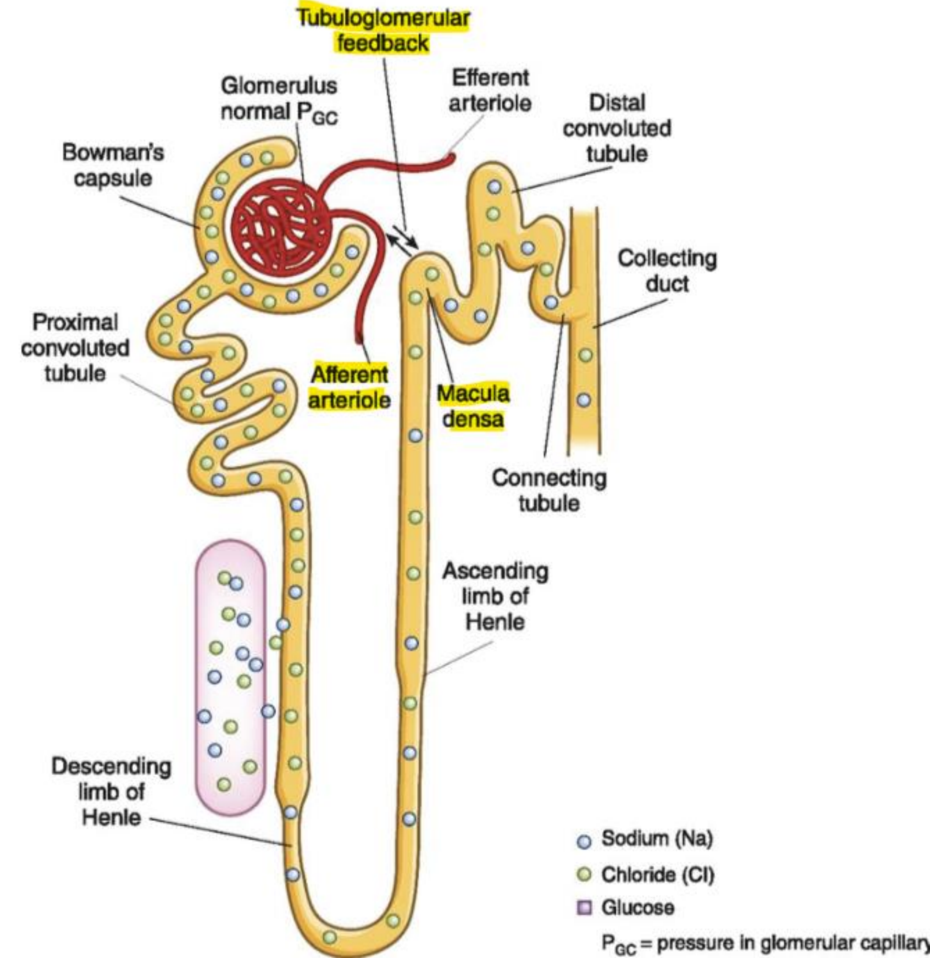
- If **macula densa** senses high NaCl/Flow (high GFR), it will tell the afferent arteriole to constrict → lower GFR back to normal
- End result: Keeps GFR Steady (Goal #1)

3) **<u>Glomerulotubular balance (GTB)</u>** (Glomerulus input → Tubule balances)

- proximal tubule always reabsorbs the **same fraction (~⅔)** of whatever is filtered — so if GFR rises 20%, reabsorption rises 20% too
- End result: Keep Na^+ delivery to distal nephron consistent (Goal #2)



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*



Change	GFR	RBF (RPF)	Filtration fraction
Afferent constriction **(less flow in)** ***Ang II (high conc.)*** ***TGF***	↓	↓	—
Efferent constriction ***Ang II (pinch the drain)***	↑	↓	↑
Afferent dilation ***PGE_2/PGI_2 (open the tap)***	↑	↑	—
Efferent dilation ***ACE-i/ARB (open the drain)***	↓	↑	↓

FF = GFR / RPF

Goal of Water Excretion (ADH)

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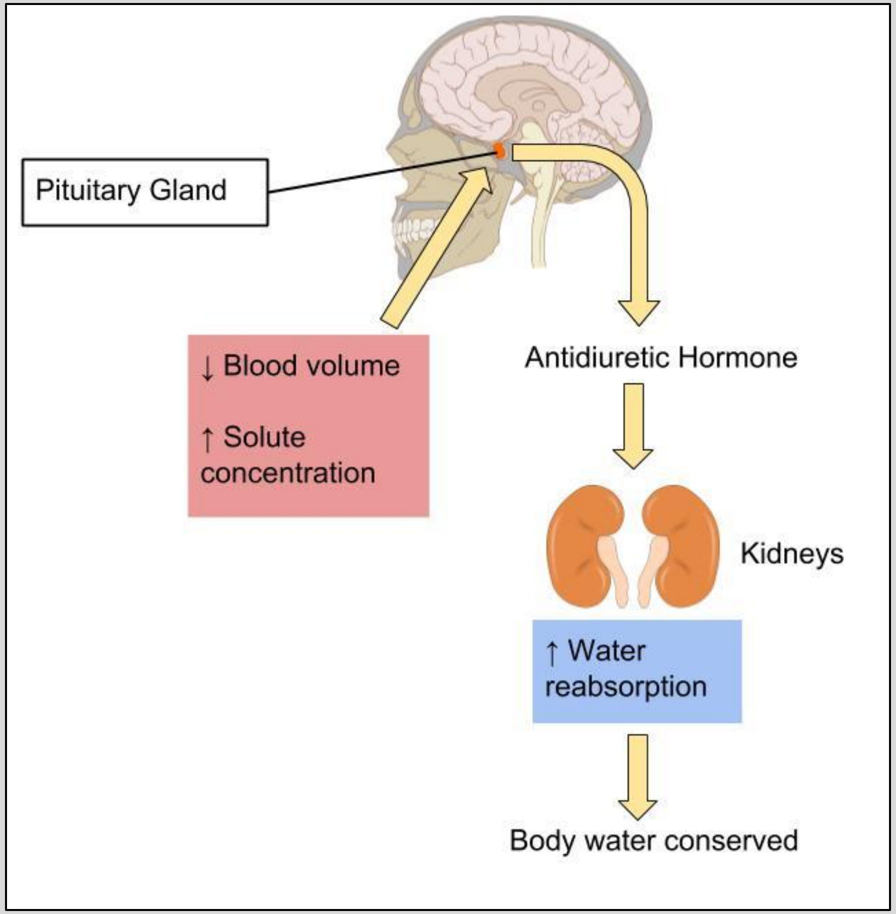
- When kidney detects abnormal **volume** → adjusts via sodium excretion (RAAS) ✓
- When kidney detects abnormal **concentration (osmolality ≈ serum [Na$^+$])** → adjusts water excretion via Antidiuretic Hormone (ADH)
- Goal of water excretion: keep plasma osmolality **(≈ serum [Na$^+$])** within homeostasis, while preserving vascular volume (set by RAAS)
- Retaining more water dilutes (↓) osmolality

(Normal serum Osm = 275 – 295)
(Normal serum [Na$^+$] = 135-145 mEq/L)



ADH = Control of Water Excretion

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Antidiuretic Hormone (ADH, Vasopressin) – produced in hypothalamic supraoptic/paraventricular nuclei → secreted via post. pituitary

- Sets how concentrated the urine is by controlling the water permeability of the collecting duct via AQP2 insertion

ADH ↑ → concentrated, low-volume urine → more water saved → serum [Na^+] falls back toward normal.

ADH ↓ → dilute, high-volume urine → water dumped → serum [Na^+] rises back toward normal.

At high levels, ADH also acts on V1 receptors as a **vasoconstrictor** (hence its alternative name, vasopressin)

too much ADH = **SIADH** → water retention → **hyponatremia**;

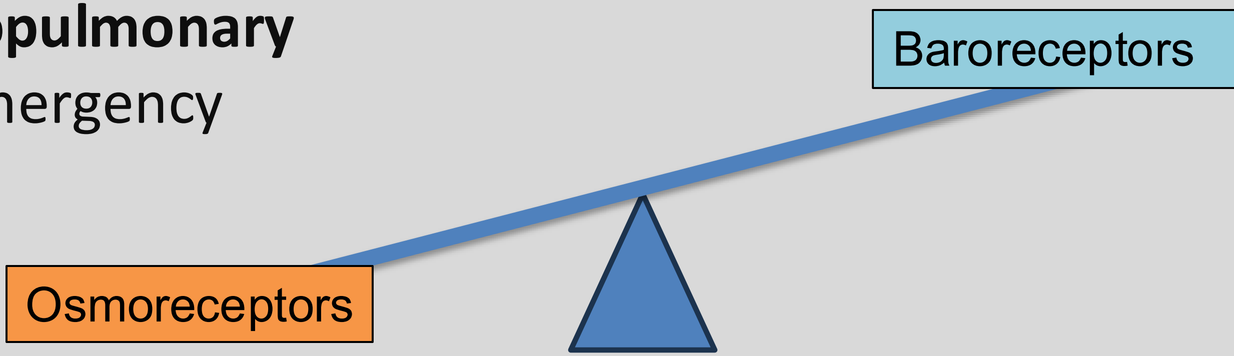
too little/ineffective ADH = **diabetes insipidus** → water loss → **hypernatremia**

Sensory Control of ADH

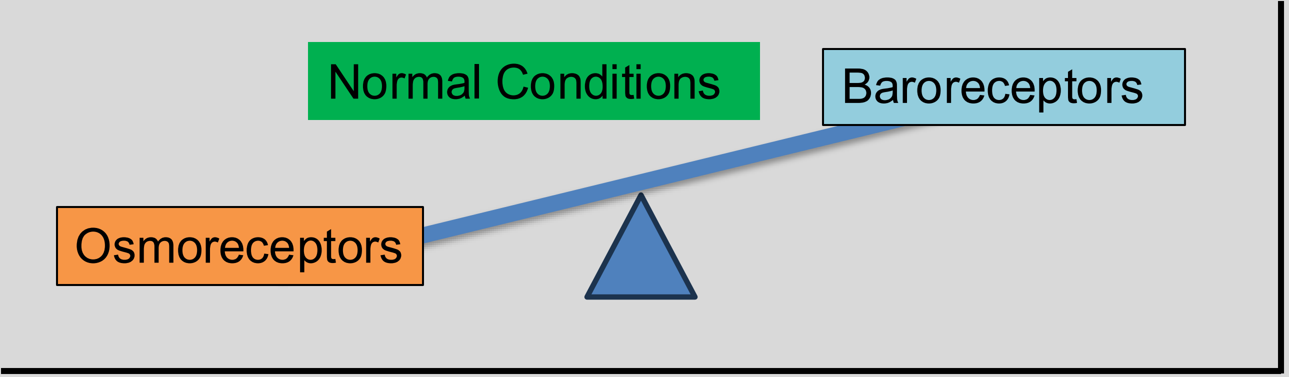
Two Main Triggers / Sensory Inputs Determining ADH release:

1) **Rising Plasma Osmolality triggers Hypothalamic Osmoreceptors** (most sensitive, day-to-day controller) → located in circumventricular organs of brain, detects rising plasma osmolality/sodium concentration/water deficit → ↑ADH → water reabsorbed → serum osmolality pulled back down

2) **Falling blood volume/pressure triggers Cardiopulmonary and Arterial Baroreceptors** (less sensitive – emergency role; see next slide)



Under normal conditions, Osmolality rules as the main trigger for ADH release.



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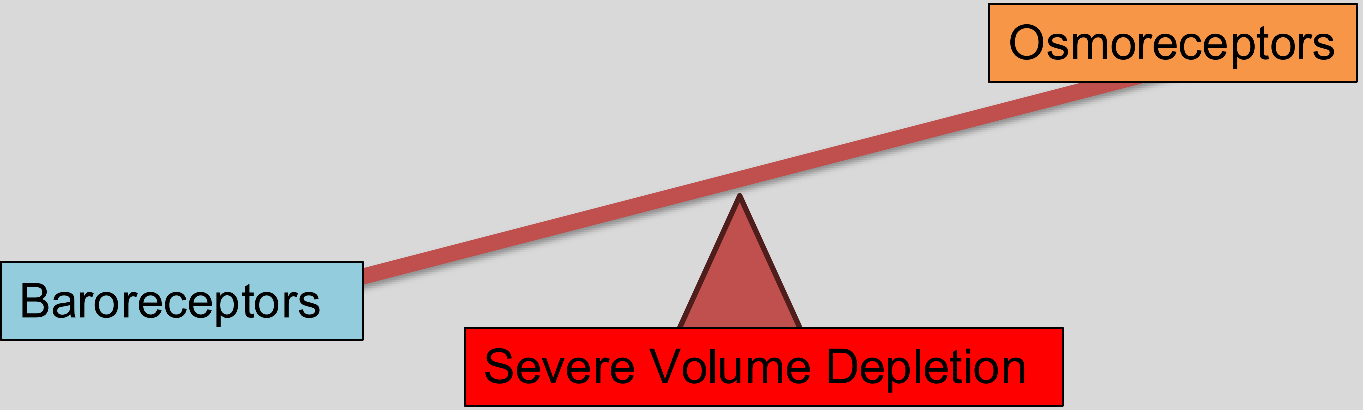

Under dangerous volume drop*: baroreceptors override the osmoreceptors → body pours out ADH even if plasma already dilute

- *This is also the case for **low effective circulating volume (ECV)** (heart failure)*

The Price of Protecting Perfusion with ADH = Hypovolemic Hyponatremia

1) High ADH release (via low volume / low ECV)

2) ↑ADH → ↑ ↑ Water retained

3) Water Retained in Excess, leading to solute dilution → [Na^+] falls causing **hyponatremia**

Additional Regulators of ADH

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- **Angiotensin II** → low volume activates RAAS, then Ang II itself stimulates ADH (pulling more water) and increases thirst/salt appetite
- Pain, severe emotional stress, nausea/vomiting = non-osmotic triggers causing ADH release
- Alcohol **inhibits** ADH

Thirst and Salt Appetite

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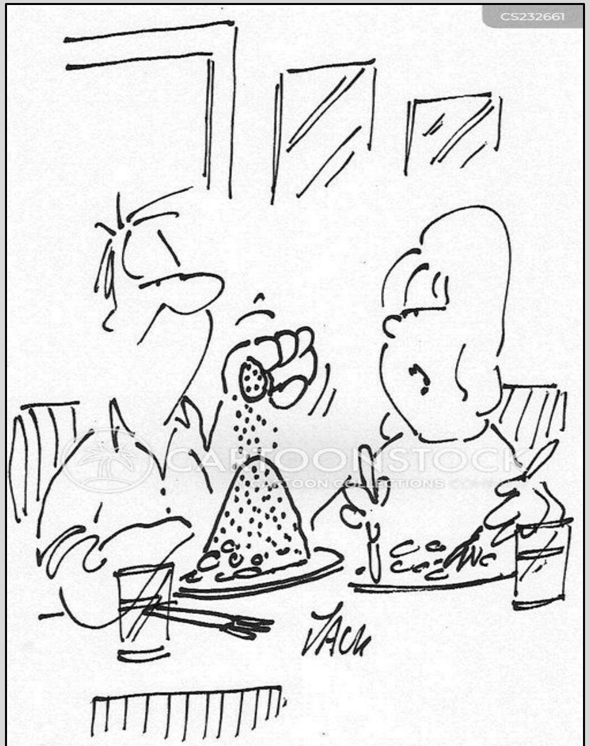
- Renal compensation alone cannot fully correct deficits of salt and water → ingestion is the ultimate compensatory mechanism

- Like ADH, stimulated and signaled by reduced plasma volume, increased serum osmolality, and Ang II

- Additionally triggered by feeling of dry mouth

- Ang II → circumventricular organs → lateral hypothalamus → thirst



"I see you are cutting back on your salt intake."

Revisiting the Three Timescales of BP Control:

Short — the baroreflex (seconds–minutes).

- Stretch receptors in the carotid sinus and aortic arch sense a pressure change and instantly adjust sympathetic outflow → heart rate, contractility, and arteriolar tone.
- This is what stops you fainting when you stand up.
- Fast, but *adapts/resets* within a day or two — can't fix a *sustained* problem.

Intermediate — RAAS / angiotensin II (minutes–hours).

- Longer-lasting drop in pressure or volume drives renin → angiotensin II, which raises pressure by **squeezing the pipes** (↑ resistance)
- Not a cure: you can't hold pressure up by vasoconstriction indefinitely: never fixes the underlying problem = the *volume* filling the vessels.

→ *(AngII's other arm = driving renal Na^+ retention via aldosterone = actually belongs to the long-term renal system below.)*

Long — the kidney (hours–days).

- Only the kidney can change the *actual amount of fluid* in the vascular tree, by adjusting how much Na^+ and water it excretes
- The **sole controller of blood pressure over the long run**
- The reason why chronic hypertension is, at its core, a kidney/volume problem.

Lecture Format

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Part 1: Where sodium and water move & why it matters

Part 2: How the body regulates those movements

Part 3: Why dysregulation causes disease: cardiovascular pathologies that involve the kidneys

Heart Failure

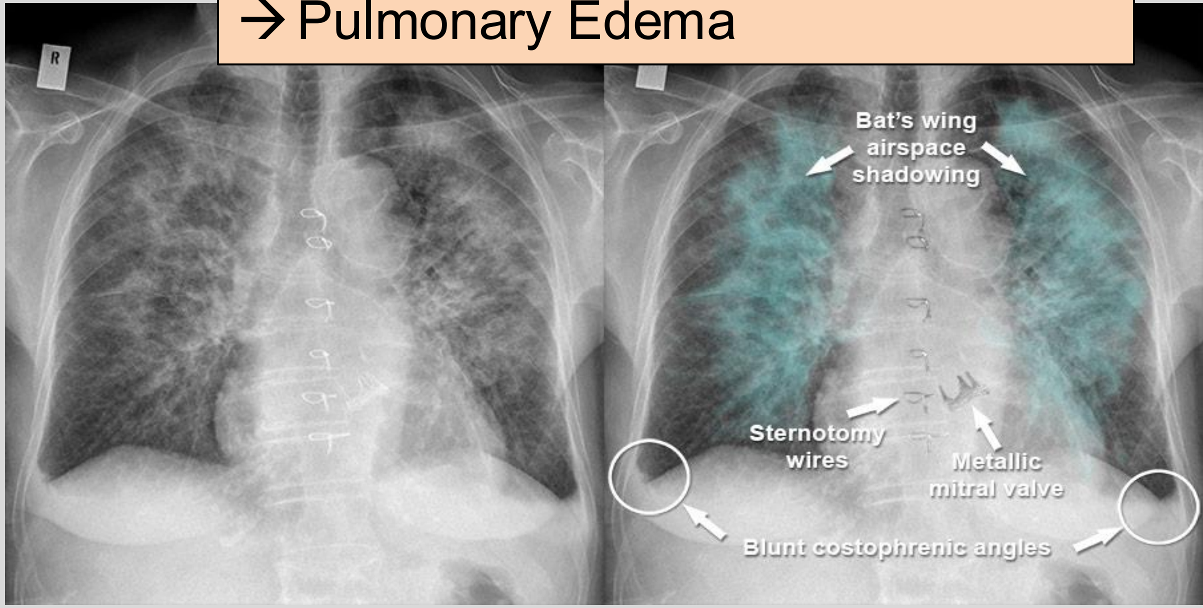
Occurs when heart muscle becomes weakened (for several reasons), resulting in a **weak pump**

1) Weak pump → ↓ Cardiac Output → less blood reaching arteries = **↓ ECV**

2) ↓ ECV perceived by renal baroreceptors as "low volume" → RAAS/sympathetics/ADH activated → kidney retains Na^+ and water → total volume raised = Edema

3) Because pump still weak, and ECV stays low, this signal never shuts off, creating a **vicious repeating cycle**

Signs of Edema in Heart Failure:
→ Pitting Edema on lower extremities
→ Ascites
→ Pulmonary Edema



Heart Failure

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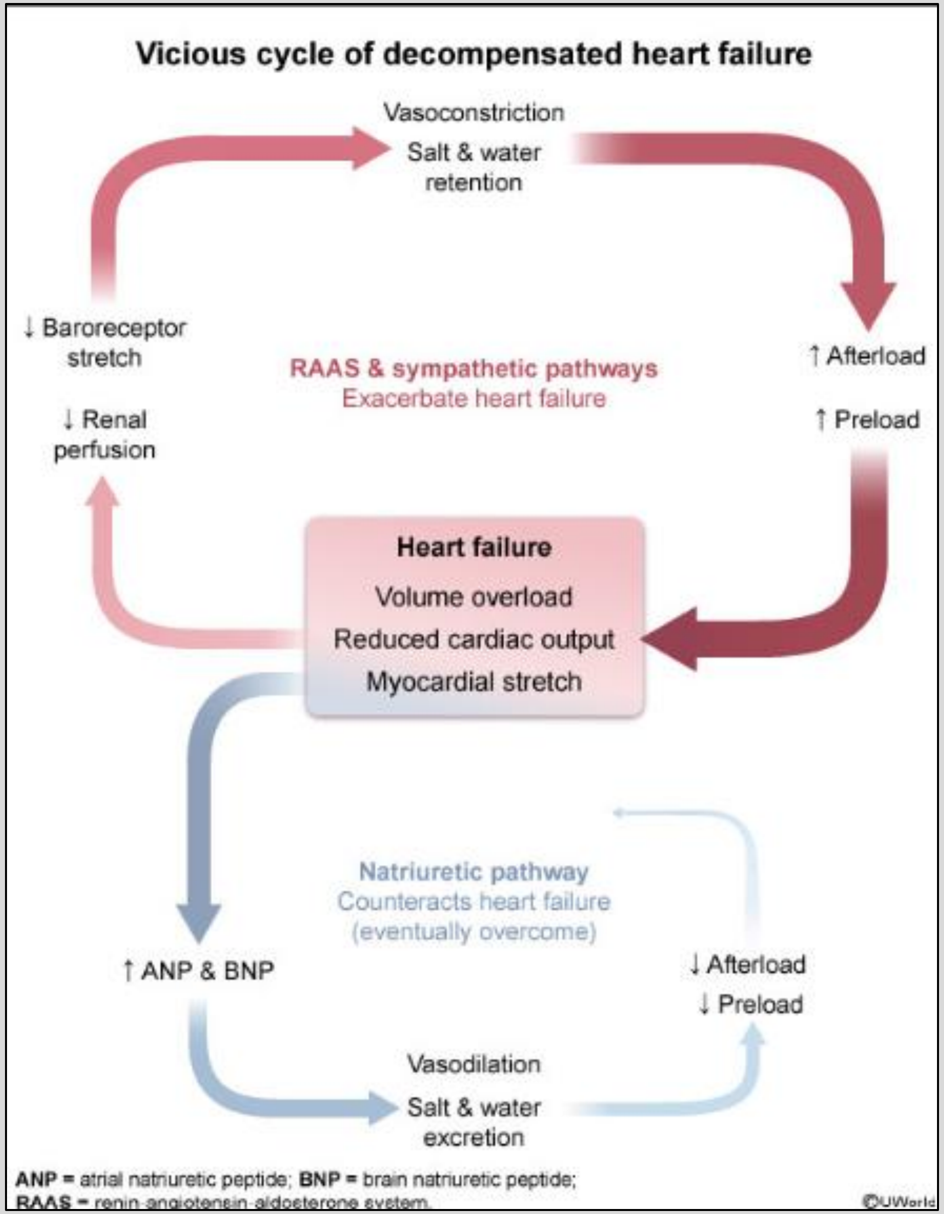
- Because heart is not pumping efficiently, excess blood pools in chambers/venous system → ↑**Natriuretic peptides (BNP)** partly oppose but can't break the cycle
  - **BNP** clinically used as a severity marker

- **Hyponatremia** occurs in severe cases due to non-osmotic release of ADH (low ECV) → a Hypervolemic Hyponatremia!!
  - Often hyponatremia in HF signifies higher risk of mortality

- Therapy targets the kidney's over-firing signals: loop diuretics (↓volume), ACE-i/ARB (↓AngII), MR antagonists (↓aldosterone), sacubitril (↑natriuretic peptides).



Returning to the Opening Question:

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A 75-year-old male with PMHx of congestive heart failure presents with complaints of dyspnea. Physical exam reveals 3+ bilateral lower extremity pitting edema (excess fluid in legs) and crackles at the bilateral lung base due to pulmonary edema (fluid buildup in lungs).

The physician mentions that his body is clearly overloaded with fluid, and CMP displays low blood sodium (128 mEq/L) (Normal = 135-145 mEq/L). The physician mentions his kidneys are avidly retaining even more salt and water, making the swelling worse.

1) If this patient is fluid overloaded, why are his kidneys acting like he's dehydrated (retaining more salt and water)?

   A = Because the kidney defends **effective circulating volume**, which is *low* in HF despite high total volume.

2) If he's retaining salt and water, why is his sodium low?

   A = **baroreceptor override** drives ADH → water retained in excess of sodium → hyponatremia

3) Why do we treat him with drugs that act on the kidney for a heart problem?

   A = symptoms causing damage are via **inappropriate signaling to the kidney:** so we switch those signals off (diuretics, ACE-i/ARB, MR antagonists)

THANK YOU ☺

Please email me with any questions: bkoflano@nyit.edu

Best of luck in your medical journeys!

Feedback:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R